

PFCMVP

Macronutrients and Micronutrients



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APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Elizabeth Boham, MD, RD, disclosed no relevant financial relationships with any ineligible companies.

Michael Stone, MD, is a consultant for Tailored Script and Calroy Health.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. List the key features and functions of these nutrient categories: proteins, fats, carbohydrates, minerals, vitamins, and phytonutrients.
2. Explain the factors that determine the nutrient density of foods.
3. Explain the factors that may impair nutrient assimilation despite adequate consumption.
4. Cite point-of-care clinical tools that assist with the education, evaluation, treatment, and tracking of patients with regard to nutrient categories.

IFM Clinical Skills and Performance Objectives

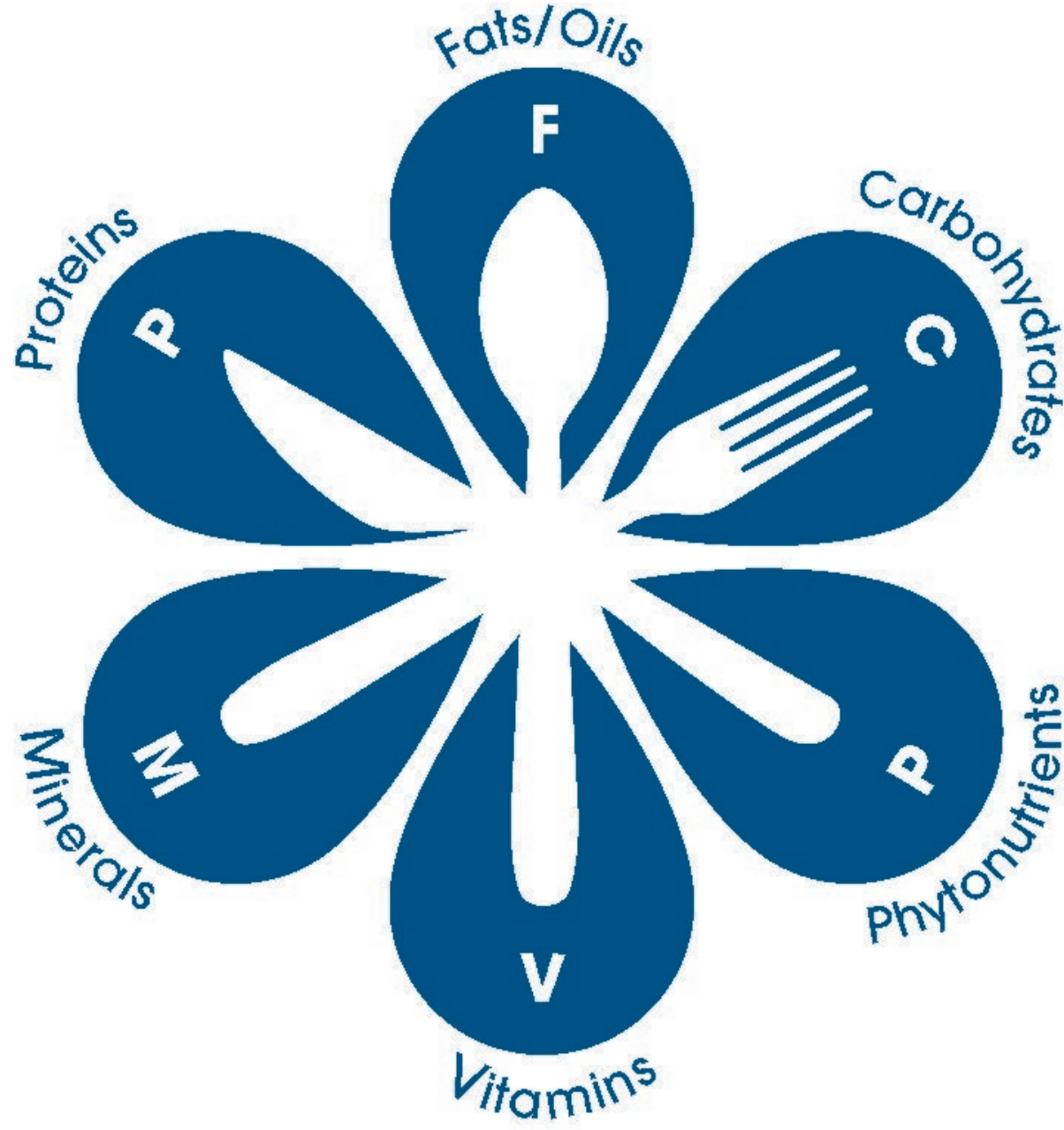
THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Recommend adequate and balanced consumption of all categories of macronutrients and micronutrients.
2. Utilize point-of-care clinical tools that assist with the education, evaluation, treatment, and tracking of patients with regard to adequacy of macronutrient and micronutrient status.

Clinical Focus

- Categories of macronutrients and micronutrients.





Macronutrients – PFC

Proteins

- Quality
- Complete amino acids
- Dispensable and indispensable
- Sulfur-containing
- Sulfur protein byproducts

Fats/Oils

- Monounsaturated (oleic)
- Essential fatty acids
- Linoleic
- Alpha linolenic
- Arachidonic
- Saturated fats
- Cholesterol
- Trans fats

Carbohydrates

- Monosaccharides
- Disaccharides
- Polysaccharides
- Fiber (soluble and insoluble)
- Artificial sugars
- High-fructose corn syrup

Micronutrients – MVP

Minerals

- Major Minerals
>100 mg/d
- Trace Minerals:
mg/day
- Ultratrace
Minerals: mcg/d

Vitamins

- Fat-Soluble
- Water-Soluble
- Fat and Water-Soluble

Phytonutrients

- Phytonutrient Spectrum:
 - White
 - Red
 - Orange
 - Yellow
 - Green
 - Blue
 - Indigo
 - Violet
 - Tan

Micronutrients – MVP

Minerals

- **Major Minerals >100 mg/d**
 - Calcium, Chloride, Magnesium, Phosphorus, Potassium, Sodium
- **Trace Minerals: mg/day**
 - Copper, Fluoride, Iron, Manganese, Selenium, Zinc
- **Ultratrace Minerals: mcg/d**
 - Chromium, Iodine, Molybdenum, Vanadium

Vitamins

- **Fat-Soluble**
 - Vitamins A, D, E (tocotrienols and tocotrienenes), K (phyloquinones K1, menaquinones K2, menadione K3), Choline
- **Water-Soluble**
 - Ascorbic Acid (C), Thiamine (B-1), Riboflavin (B-2), Niacin (B-3), Pantothenic Acid, Biotin, Folate (B-9), Cobalamin (B-12)
- **Fat and Water-Soluble**
 - Lipoic Acid

Phytonutrients

- **Phytonutrient Spectrum:**
 - White
 - Red
 - Orange
 - Yellow
 - Green
 - Blue
 - Indigo
 - Violet
 - Tan
- **Bioflavonoid Examples:**
 - Quercetin, Rutin, Hesperidin, Anthocyanins, Lutein

Clinical Focus

- PFCMVP – Protein



Macronutrients



Protein

Structural Support and Functional Roles

Structural Support

- Muscle
- Skin
- Hair
- Bone
- All connective tissue

Functional Roles

- Hormones
- DNA
- Neurotransmitters
- Tissue repair and growth
- Toxin biotransformation
- Blood oxygenation
- Myriad cellular processes

Protein

Food Sources (and sourcing of foods)

Animal
Sources
(Complete)

- Meat
- Poultry
- Fish
- Eggs
- Shellfish/seafood
- Dairy

Plant
Sources
(Incomplete)

- Deep leafy greens
- Cruciferous vegetables
- Beans/legumes
- Nuts/seeds

Toolkit – Macronutrients: Proteins

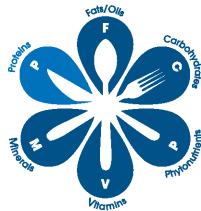


Macronutrients:

Proteins

Proteins are found in every living thing on earth. In the human body, protein is an essential component of muscles, skin, hair, and bones, and is found in nearly every other tissue and body part. Protein is essential for proper muscle development and function, bone health, connective tissue strength, tissue repair and growth, blood oxygenation, and basic cell activity.

All proteins are made up of amino acids, which are considered to be the building blocks of life. Our hormones, DNA, neurotransmitters, and most of our muscles are made almost entirely from amino acids. Our bodies do not store protein, nor do they make all of the amino acids needed for the body to function properly. This means that we must take in protein from our food on a daily basis. Thankfully, protein is found in many different foods.



Macronutrients are a class of compounds that provide humans with energy and essential nutrients. They are required by the body in relatively large amounts on a daily basis and make up the bulk of the diet. Proteins, fats, and carbohydrates are the main types of macronutrients that provide the body with energy.

Sources of Protein

Animal Proteins: Proteins from animal sources are considered to be “complete” proteins, because they contain all of the amino acids your body needs to get from food (known collectively as essential amino acids). Animal proteins include meat, poultry, eggs, fish, and shellfish.

- **Dairy products:** Dairy products like milk and yogurt are good sources of both protein and carbohydrates. Cheese is also a good protein source, but is low in carbohydrates. Note that not everyone can digest dairy properly, and dairy is not necessary to include in a balanced diet. For those who choose to consume dairy, organic sources are preferred.

Plant Proteins: Most plant proteins are considered to be “incomplete,” because they contain low levels of some essential amino acids. High-protein vegetables include dark leafy greens (collard greens, spinach, mustard greens), asparagus, bok choy, broccoli, Brussels sprouts, and cauliflower. Foods like tofu, tempeh, and spirulina are also plant proteins.

- **Beans and legumes:** Beans and legumes are good sources of both protein and carbohydrates. They also contain other vitamins and minerals like folate (vitamin B9), potassium, iron, and magnesium. Combining beans and legumes with whole grains (not necessarily at the same meal) supplies the body with all essential amino acids, forming a complete protein.
- **Nuts and seeds:** Nuts and seeds are sources of both protein and fat. They also contain many other vitamins and minerals like magnesium, zinc, selenium, and copper.

Recommendations for Protein Intake

In the United States, the recommended dietary allowance of protein is 46 grams per day for women and 56 grams per day for men. These figures apply to healthy adults over the age of 19, and are a measure of the level of protein needed to meet daily nutritional requirements. Other research suggests that adults should get a minimum of 8 grams of protein for every 20 pounds of body weight.

According to the Dietary Guidelines for Americans, protein should account for about 10% to 35% of calories per day. Extra protein is required for pregnant or nursing women, athletes, active individuals, the elderly, and those suffering from or recovering from an illness. IFM recommends that protein should account for 20% to 30% of daily calories for most individuals, including those with health concerns requiring advanced therapeutic dietary interventions.

If you're unsure about how much protein you should be eating, ask your functional medicine practitioner. You may need extra protein to help build or repair tissues temporarily, but remember that excess protein you consume is not stored in the body. Consuming too much protein on a regular basis can put a strain on your kidneys, which can lead to serious health conditions.

Tips for Incorporating Protein Into a Balanced Diet

- Eat some protein with each meal and snack in order to distribute protein evenly throughout the day.
- Replace instant oatmeal with a vegetable omelet or Greek yogurt topped with berries and almond slices.
- Choose non-dairy milks that contain protein, such as soy milk or oat milk.
- Try edamame, roasted chickpeas, or a bean dip with vegetables for an afternoon snack.
- Add beans or legumes to a salad, such as chickpeas or lentils.



REFERENCES

1. U.S. Department of Agriculture and U.S. Department of Health and Human Services. Dietary Guidelines for Americans, 2020-2025, 9th Edition. December 2020. Available at [DietaryGuidelines.gov](https://www.dietaryguidelines.gov).

Toolkit Item – Food Sources: Lean Protein



Food Sources:

Lean Protein

Protein is an essential nutrient needed for the growth, repair, and support of body tissues. Protein is also needed to make hormones, enzymes, and immune cells.

Most people should have a small amount of protein with every meal and snack to help regulate blood sugar and promote satiety. The benefits of protein come from amino acids, the building blocks of protein.

There are nine essential amino acids, which can be obtained from plant or animal protein.^{1,2} Below are guidelines for choosing both types of protein. For personalized protein recommendations, talk to a functional medicine clinician.

Lean and Clean

When choosing proteins, “lean” and “clean” are two guiding principles to follow. The **lean principle** means selecting proteins with little-to-no saturated fat, such as beans or skinless chicken breasts. Choosing lean proteins is important for individuals who need to limit their fat intake because of high cholesterol or heart disease.

In following the **clean principle**, it is strongly advised to purchase organic, non-GMO (non-genetically modified) products for all protein sources to avoid herbicides, pesticides, and other chemicals. Reducing exposure to these chemicals is important for many aspects of health.

Plant Proteins

- **Grains:** quinoa, amaranth
- **Legumes:** dried beans, peas, lentils, and pasta made from these
- **Nuts:** unsalted varieties of nuts, including almonds, Brazil nuts, cashews, hazelnuts, macadamias, peanuts, pecans, pine nuts, pistachios, walnuts, and butters made from these
- **Seeds:** pumpkin seeds, sesame seeds, sunflower seeds, chia seeds, flaxseeds, and butters made from these
- **Soy:** tofu, tempeh, soy yogurt, and “meat” alternatives*

**Some meat alternatives are highly processed and contain a lot of salt and fat. Read food labels and select products with a minimal number of ingredients.*



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Animal Proteins

- **Beef or bison (grass-fed preferred):** brisket, chuck arm, flank steak, bottom round, eye of the round, top round, sirloin, tenderloin, tip roast, top loin
- **Dairy:** low-fat cheese, fat-free cottage cheese, skim milk, fat-free or Greek yogurt
- **Eggs:** egg whites, hard-boiled eggs
- **Fish (wild-caught preferred):** cod, flounder, haddock, halibut, mahi-mahi, perch, pollock, salmon (chum, pink), sole, snapper, tilapia³
- **Lamb:** blade chop, foreshank, leg roast (shank half), loin chop, sirloin roast
- **Pork:** center cut chop, center-cut loin roast, ham (90-95% lean), leg (shank half), tenderloin, Canadian bacon
- **Poultry:** skinless chicken or turkey
- **Seafood:** clams, crab, lobster, oysters, scallops, shrimp



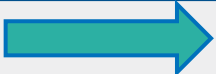
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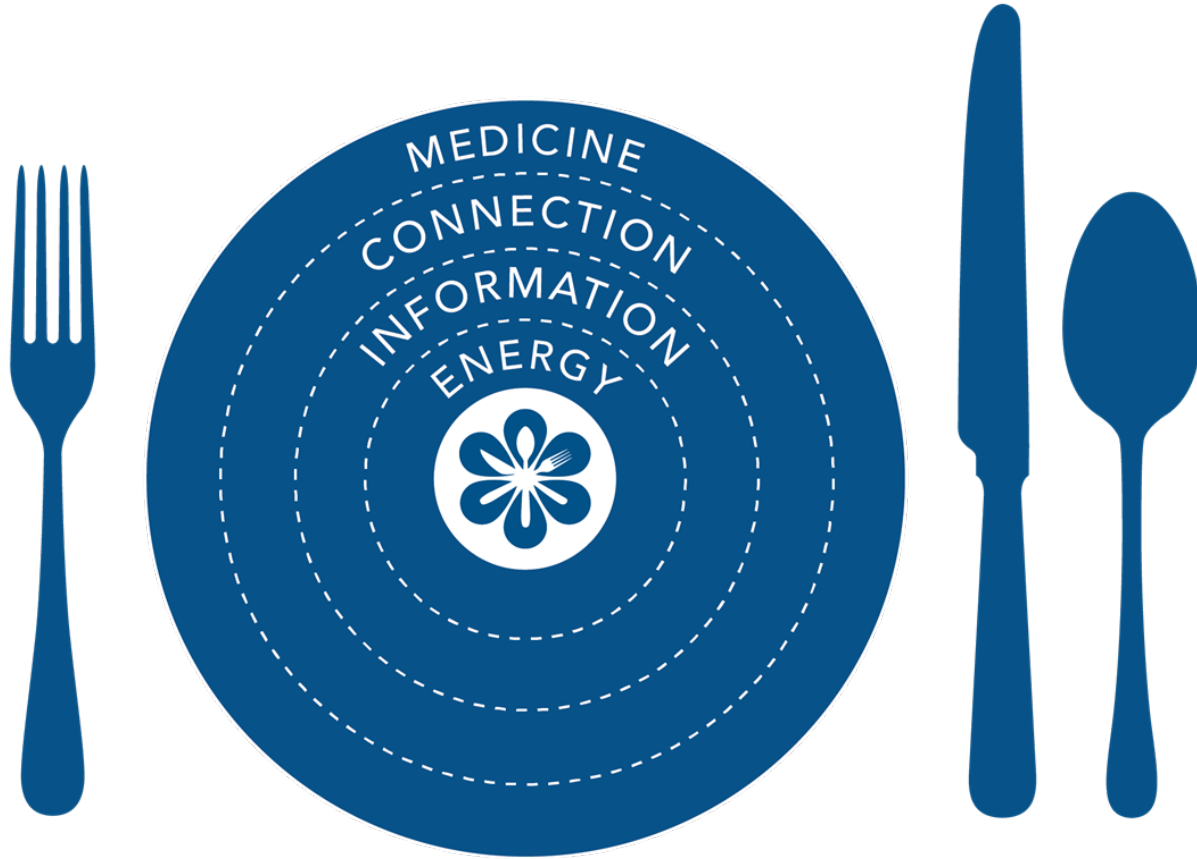
Daily Protein Requirements

154 lb/70 kg

Authority	Recommendation (g/kg/day)	Protein/Day (g)
Food and Nutrition Board	 0.8	56
Elango R, Humayun M (amino acid oxidation technique)	0.93 and 1.2 (41 and 50%, respectively, higher than the DRI recommendations)	65-84
International Society of Sports Nutrition	1.4-2.0	98-140
	Sarcopenia: 1.6	

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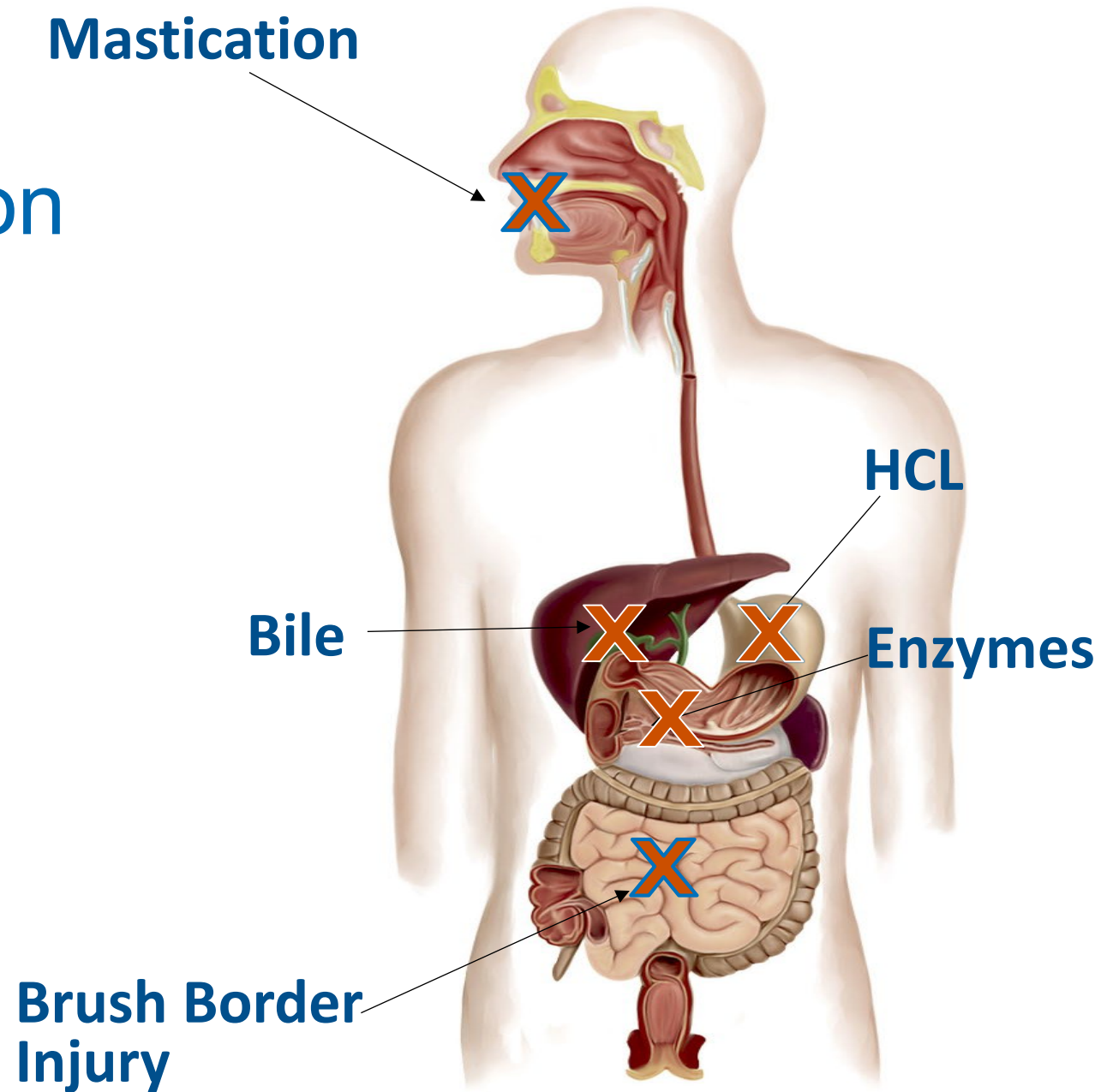
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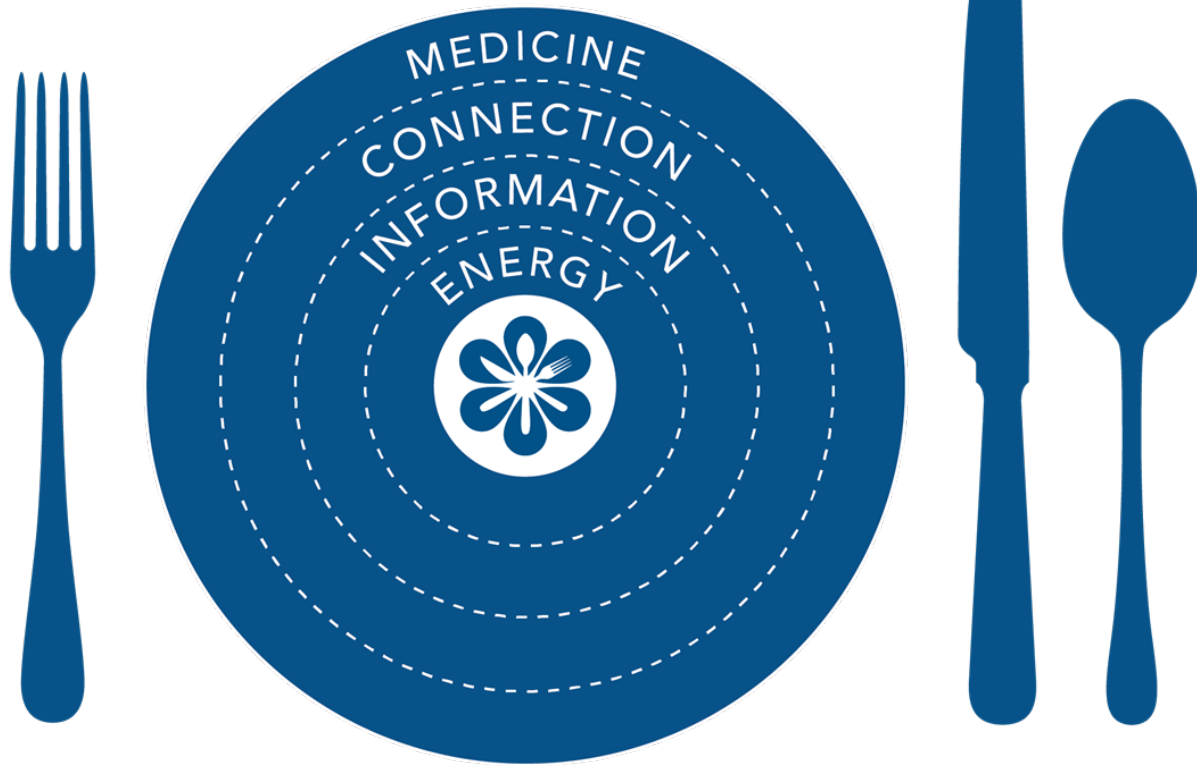
- What we eat
- What we digest
- What we absorb
- What we assimilate

Impairments in Protein Digestion and Absorption

- Inadequate mastication
- Hypochlorhydria (esp. PPIs)
- Pancreatic insufficiency
- Brush border injury
- Dysmotility/diarrhea
- Dysbiosis and GI infections
- IBD



We are...



- What we need
- Consider increased protein requirements for certain individuals:
 - Pregnant/nursing
 - Elderly
 - Excessive exercise
 - Elite athletes
 - Elevated toxic load

Clinical Focus

- PFCMVP – Fats and Oils



Macronutrients



Fats and Oils

Structural Support and Functional Roles

Structural Support

- Brain
- Cellular membranes

Functional Roles

- Energy source
- Energy storage

Fats and Oils

Food Sources (and sourcing of foods)

Fats and
Oils to
Favor

- Mono-unsaturated
- Poly-unsaturated
- Omega-3

Fats and
Oils to
Limit or
Avoid

- Trans
- (Partially) hydrogenated
- Saturated
- Omega-6

Toolkit – Macronutrients: Fats and Oils



Macronutrients are a class of compounds that provide humans with energy and essential nutrients. They are required by the body in relatively large amounts on a daily basis and make up the bulk of the diet. Proteins, fats, and carbohydrates are the main types of macronutrients that provide the body with energy.

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Macronutrients:

Fats and Oils

Fats and oils provide a lot of energy (in the form of calories) in a relatively small amount of food. Once eaten, fat can be stored by the body for later use. When food is scarce, this stored fat becomes a source of fuel. Fat also provides necessary support for the brain, connective tissue, and digestive system. In short, fat plays an important role in health, and is an essential component of a balanced diet.

Different Types of Fat and Their Sources

The list below explains the various types of fats and foods that contain them. Keep in mind that all fat-containing foods contain a mix of different types of fats. In general, you should avoid trans fat and limit saturated fats in your diet. In contrast, unsaturated fats are generally considered healthy when their food sources are whole, natural foods.

1. Trans Fat: This type of fat is created by food manufacturers to transform a liquid to a solid in order to extend the shelf life of a food product. The FDA has largely banned artificial trans fat (in the form of partially hydrogenated oils) from the food supply because of its harmful effect on heart health. Despite this ban, exceptions were made to allow artificial trans fat to remain in certain processed foods like coffee creamers, frozen desserts, fried foods, and microwave popcorn. Trans fat also naturally occurs in red meat and dairy products, though the health impact of natural trans fat is less established than artificial trans fat. It is strongly recommended to **avoid this type of fat** entirely. To avoid trans fat, check the ingredient list of foods to make sure there are no “partially hydrogenated” oils included.

2. Saturated Fat: This type of fat is solid at room temperature. It is found in animal proteins (like beef, pork, and chicken) and dairy products. It is also found in some plant-based fats and oils, including coconut and coconut oil, palm oil, palm kernel oil, and hydrogenated oils. Coconut oil contains a unique type of healthy fat, called medium chain triglycerides, however it should be eaten in moderation like other sources of saturated fat.

When trans fats were banned by the FDA, many food manufacturers replaced trans fat with saturated fat (in the form of fully hydrogenated oils). In the U.S., saturated fat in the diet comes mostly from dairy products (such as cheese and ice cream), burgers, sandwiches, pizza, desserts, baked goods, and pasta. It is recommended to **limit this type of fat** to no more than 7% to 10% of calories each day, and to opt for healthier unsaturated fats when possible.

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3. Unsaturated Fats: These healthy fats are liquid at room temperature. They are heart-healthy, and they can help reduce inflammation. They are mostly found in plant foods like vegetable oils, nuts, and seeds. There are several types of unsaturated fats:

- **Monounsaturated fats:** This type of unsaturated fat has one double bond in its chemical structure. These fats can improve blood cholesterol levels, and they also help keep the body's cells healthy. Good sources of monounsaturated fats are avocados and avocado oil, olives and olive oil, almonds, hazelnuts, pecans, pumpkin seeds, and sesame seeds.
- **Polyunsaturated fats:** This type of unsaturated fat has more than one double bond in its chemical structure. Sources of polyunsaturated fats include fish, vegetable oils, and seeds:
 - **Omega-3 fatty acids:** These polyunsaturated fatty acids are essential. This means that the body can't make them, so we must get them from food. Omega-3s are important for brain health, development, and reducing inflammation, among other things. Good sources of omega-3s are fatty fish (especially herring, oysters, salmon, and sardines), flaxseeds and flaxseed oil, chia seeds, and walnuts.
 - **Omega-6 fatty acids:** These polyunsaturated fatty acids are also essential, so we must get them from food. However, our diets are often too high in omega-6s from processed foods, which cause inflammation. Processed vegetable oils (like corn, sunflower, and soybean oil) are high in omega-6s, as are foods fried in or made with these oils (such as tortilla chips or baked goods). Omega-6s are also found in healthy foods like nuts and seeds (almonds, cashews, walnuts, and sunflower seeds).

Recommendations for Fat Intake

According to the Dietary Guidelines for Americans, adults should get 20% to 35% of their calories from fat. Infants and young children need a slightly higher proportion of fat (25% to 40%) for proper brain development. IFM recommends that fat should account for about 30% of daily calories for most healthy individuals. For those with specific health concerns requiring advanced therapeutic dietary interventions, fat recommendations may range from roughly 30% to 60% of daily calories, depending on the specific intervention or health condition. Talk to your functional medicine provider to learn what recommendation is best for you.

Tips for Incorporating Fats and Oils into a Balanced Diet

- Top your salad with healthy oils (like extra virgin olive or walnut) instead of store-bought, shelf-stable salad dressings. Add a squeeze of lemon and balsamic vinegar for flavor.
- Incorporate fatty fish, like salmon, into your meals twice a week.
- Add freshly-ground flaxseeds or flaxseed oil to your morning smoothie or oatmeal.
- Have a handful of raw, unsalted nuts as a snack.
- Add avocado slices to sandwiches or salads. It can be a delicious, heart-healthy substitute for mayonnaise.

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1. U.S. Department of Agriculture and U.S. Department of Health and Human Services. Dietary Guidelines for Americans, 2020-2025. 9th Edition. December 2020. Available at [DietaryGuidelines.gov](https://www.dietaryguidelines.gov).

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Medicine

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Toolkit Item – Food Sources: Essential Fatty Acids



The Adequate Intakes (AIs) for omega-6 fatty acids are as follows:

- Females, ages 19-50: 12 grams per day
- Females, ages 51+: 11 grams per day
- Pregnancy, lactating: 13 grams per day
- Males, ages 19-50: 17 grams per day
- Males, ages 51+: 14 grams per day

The Adequate Intakes (AIs) for omega-3 fatty acids are as follows:

- Females, ages 19+: 1.1 grams per day
- Pregnancy: 1.4 grams per day
- Lactating: 1.3 grams per day
- Males, ages 19+: 1.6 grams per day



Food Sources:

Essential Fatty Acids

Essential Fatty Acids (EFAs) cannot be created by your body. They are called “essential” because you must get them through your diet. There are two primary families of essential fatty acids: omega-6 and omega-3.

Linoleic acid (LA) and arachidonic acid (AA) are two common omega-6 fatty acids. LA is found in most plant oils, and AA is present in meat, poultry, and eggs. Alpha-linolenic acid (ALA), eicosapentaenoic acid (EPA), and docosahexaenoic acid (DHA) are the most common omega-3 fatty acids. Some plant foods contain ALA, while fatty fish are the best source of EPA and DHA.

Every cell in the body needs EFAs for proper structure and function. EFAs promote healthy nerve functioning and play a role in inflammation.

Food, Standard Serving Size	Amount of LA Omega-6s (in grams)
Walnuts, 1 ounce (14 halves)	10.8
Safflower oil, 1 Tbsp	10.1
Sunflower seeds (oil roasted), 1 ounce (¼ cup)	9.7
Grapeseed oil, 1 Tbsp	9.5
Pine nuts, 1 ounce (¼ cup)	9.4
Sunflower oil, 1 Tbsp	8.9
Hemp seeds (hulled), 3 Tbsp	8.3
Pecans (oil roasted), 1 ounce (19 halves)	6.4
Pumpkin seeds (dried), 1 ounce (¼ cup)	5.9
Sesame oil, 1 Tbsp	5.6

Food, Standard Serving Size	Amount of Omega-3s (in grams)
Flaxseed oil, 1 Tbsp	7.3 (ALA)
Chia seeds, 1 ounce (2 Tbsp)	5.1 (ALA)
Hemp seeds (hulled), 3 Tbsp	2.6 (ALA)
Walnuts, 1 ounce (14 halves)	2.6 (ALA)
Flaxseed (whole), 1 Tbsp	2.4 (ALA)
Atlantic salmon (cooked), 3 ounces	1.8 (EPA and DHA)
Atlantic herring (cooked), 3 ounces	1.7 (EPA and DHA)
Sardines (canned), 3 ounces	1.2 (EPA and DHA)
Atlantic mackerel (cooked), 3 ounces	1.0 (EPA and DHA)

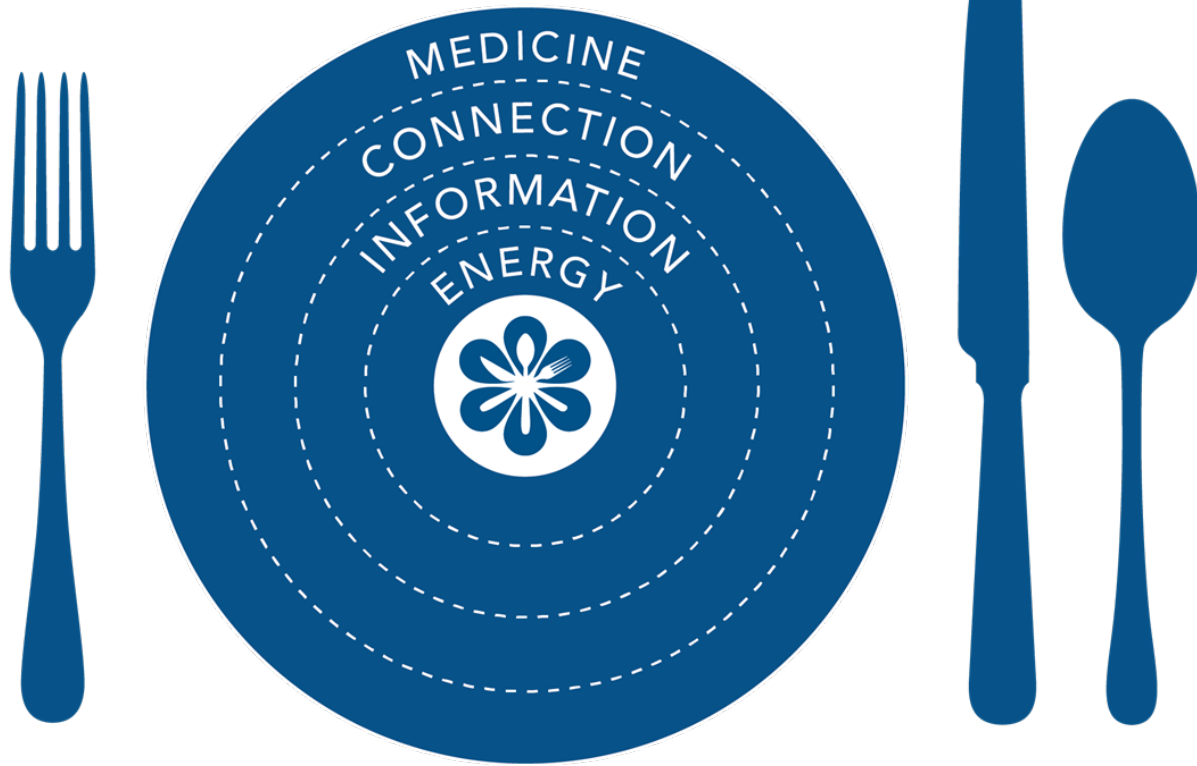
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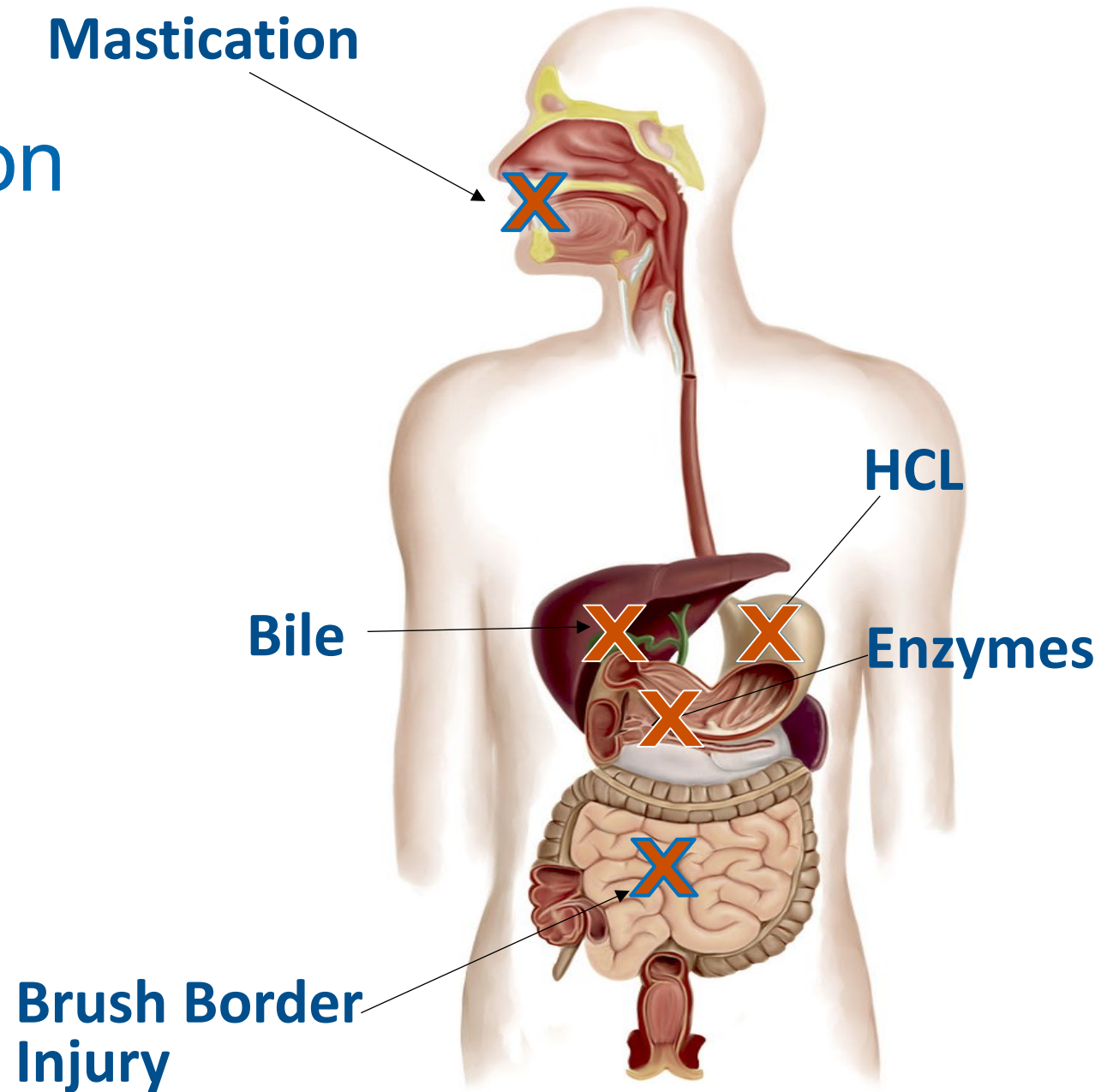
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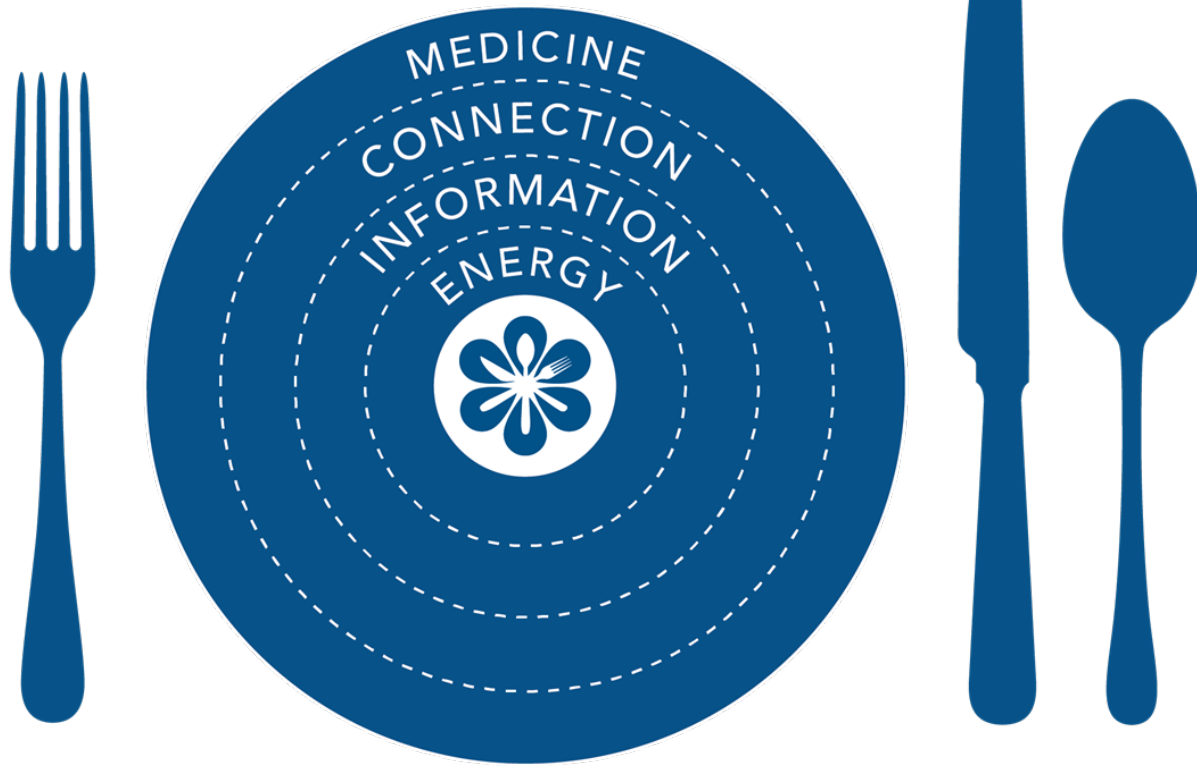
- What we eat
- What we eat ate
- What we digest
- What we absorb
- What we assimilate

Impairments in Fat Digestion and Absorption

- Bile acid disorders
- Liver or gallbladder dysfunction
- Post-cholecystectomy
- Brush border injury
- Dysmotility/diarrhea
- Dysbiosis and GI infections
- IBD



We are...



- What we need
- Consider increased fat requirements for certain individuals:
 - Children
 - Pregnant women
 - Genetic uniqueness

Clinical Focus

- PFCMVP – Carbohydrates



Macronutrients



Carbohydrates

Structural Support and Functional Roles

Structural Support

- Cell membranes

Functional Roles

- Energy stores
- Energy source
- Micronutrient source
- Gut motility (fiber)

Carbohydrates

Food Sources (and sourcing of foods)

Carbohydrates
to Favor

- Vegetables
- Fruits
- Whole grains
- Beans/legumes
- Dairy

Carbohydrates
to Limit or
Avoid

- Refined
- Ultra-processed

Toolkit – Macronutrients: Carbohydrates



Macronutrients:

Carbohydrates

Carbohydrates are the body's main source of energy. When you eat carbohydrates, your body uses some of them for energy right away. The carbohydrates that your body doesn't need are stored either in muscle cells or fat cells to be used later. Carbohydrates are found in many different foods and are an important part of a healthy balanced diet, but the source and quality of carbohydrates is important to consider.



Macronutrients are a class of compounds that provide humans with energy and essential nutrients. They are required by the body in relatively large amounts on a daily basis and make up the bulk of the diet. Proteins, fats, and carbohydrates are the main types of macronutrients that provide the body with energy.

Sources of Carbohydrates

Some carbohydrate-rich foods are more nutrient dense than others. The best sources of carbohydrates supply the body with energy while also providing it with vitamins, minerals, phytonutrients, and fiber. All of these nutrients are important for basic brain functioning, kidney health, digestive health, and muscle function. Good sources of carbohydrates include the following:

- **Vegetables:** Both starchy and non-starchy vegetables are good sources of carbohydrates. Vegetables that are high in fiber can help stabilize blood sugar and keep you feeling full for longer. To learn more about the role of fiber in a balanced diet, see IFM's [What is Fiber?](#) document.
- **Fruits:** Many fruits are high in carbohydrates due to the natural sugars found in them. Fruit is also an important source of many vitamins, minerals, fiber, and phytonutrients, and should be included in limited amounts in a balanced diet.
- **Whole Grains:** Whole, unprocessed grains are high in fiber, slow to digest, and full of vitamins and minerals. Because they are so nutrient-dense, they promote feelings of fullness, keeping you satisfied for longer.
- **Beans and Legumes:** Beans and legumes are good sources of both carbohydrates and protein. They also contain fiber and other vitamins and minerals like folate (vitamin B9), potassium, iron, and magnesium.
- **Dairy Products:** Dairy products like milk, yogurt, and cheese are good sources of both carbohydrates and protein. While they don't contain fiber, they do supply the body with other vitamins and minerals. Note that not everyone can digest dairy properly, and dairy is not necessary to include in a balanced diet. For those who choose to consume dairy, organic sources are preferred.

Other carbohydrate-rich foods include foods that are made with refined grains and sugars. This list includes (but is not limited to) sodas, white breads, pastas, cakes, pastries, cookies, chips, French fries, candy, ice cream, and

other convenience foods. While it may be unrealistic to avoid these foods at all times, too many refined grains and sugars in the diet can promote chronic inflammation in the body, which can contribute to a host of health conditions. Health experts agree that less than 10% of your daily calories should come from added sugars.

All carbohydrates are not equal in terms of how easily they are absorbed. Carbohydrates that are absorbed quickly can cause spikes in blood sugar and have negative effects. While processed foods are generally the worst offenders, even whole foods have varying effects on blood sugar. For more information about carbohydrates and how they can affect the body, see IFM's [Glycemic Index and Glycemic Load](#) document.

Recommendations for Carbohydrate Intake

According to the Dietary Guidelines for Americans, adults should get 45% to 65% of their calories from carbohydrates. However, too many carbohydrates in the diet—especially those from refined sugars and grains—can contribute to the development of many chronic health conditions. Because of this, IFM recommends that carbohydrates should account for 45% to 50% of daily calories for most healthy individuals. For those with specific health concerns requiring advanced therapeutic dietary interventions, carbohydrate recommendations may be adjusted to roughly 20% to 40% of daily calories, depending on the health condition. Talk to your functional medicine provider to learn what recommendation is best for you.

Tips for Incorporating Carbohydrates into a Balanced Diet

- Try adding whole, unprocessed grains to salads, soups, and stews. Experiment with your favorite recipes using grains like amaranth, quinoa, millet, barley, and wild rice.
- Make your favorite recipes using beans or legumes in place of animal protein. Chickpeas, black beans, tofu, and edamame work well in many dishes.
- Replace instant oatmeal and cold cereals (which usually contain added sugars) with plain full-flake or steel cut oats. Top with fresh berries for a bit of nutrient-dense sweetness.
- When shopping for bread and pastas, look for options that are made from 100% whole grains.
- Opt for whole fruit over juice when possible. Whole fruits contain more fiber and less sugar, which helps keep blood sugar balanced.



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Toolkit Item – Food Sources: Dietary Fiber



Food Sources:

Dietary Fiber

Dietary fiber comes from plant foods, including fruits, vegetables, legumes, nuts, seeds, and grains. The fiber in plant foods is not digested by enzymes present in the digestive tract, but it may be digested by the microorganisms that inhabit the intestines.

Dietary fiber is usually described as “soluble” or “insoluble,” based on its ability to dissolve in water. For example, the inner portion of an apple contains soluble fiber, whereas the peel is made of insoluble fiber. Soluble fiber contributes to a feeling of fullness and helps maintain a healthy weight. It also decreases the absorption of dietary sugars and fats, thereby helping to manage blood sugar and blood fat levels. Soluble fiber also serves as a food source for the beneficial bacteria that inhabit the digestive tract. The insoluble fiber in plant foods is helpful in moving waste products through the digestive tract. It also provides bulk to the stool and is beneficial in preventing constipation, hemorrhoids, and diverticuli.

The Dietary Reference Intakes for dietary fiber (soluble and insoluble fiber, combined) are as follows:

- Females, age 18-50: 25 grams per day
- Females, ages 51 and above: 21 grams per day
- Males, ages 18-50: 38 grams per day
- Males, ages 51 and above: 30 grams per day

Food Sources of Soluble Fiber (food, standard serving size)	Amount of Dietary Fiber (in grams)
Oat bran, ½ cup	7.0
Black beans (cooked), ¼ cup	5.4
Lima beans, ¼ cup	5.3
Tofu, ¼ cup	2.8
Avocado, ½ whole	2.1
Brussels sprouts, ½ cup	2.0
Sweet potato (cooked), ½ cup	1.8
Asparagus (cooked), ½ cup	1.7

Food Sources of Insoluble Fiber (food, standard serving size)	Amount of Dietary Fiber (in grams)
Wheat bran, ½ cup	12.5
Navy beans (cooked), ½ cup	9.5
Kidney beans (cooked), ½ cup	8.2
Lentils (cooked), ½ cup	7.8
Black beans (cooked), ½ cup	7.5
Okra, ½ cup	3.1
Turnip (cooked), ½ cup	3.1
Peas, ½ cup	3.0

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3. Oregon State University, Linus Pauling Institute. Micronutrient Information Center: Fiber. <https://lpi.oregonstate.edu/mic/other-nutrients/fiber>. Updated June 2019. Accessed March 23, 2020.

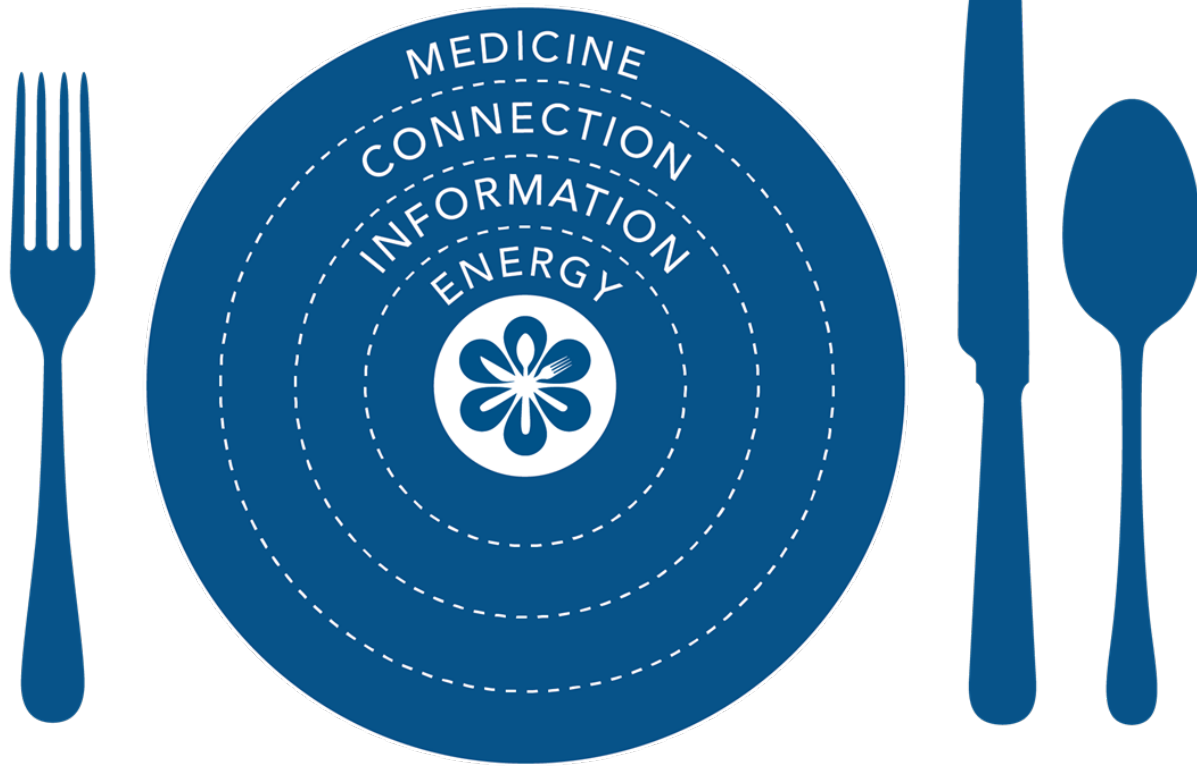


Carbohydrates and Glycemic Load

Individual Variability – Quality and Quantity Matters

- How much carbohydrate is consumed?
- How fast is it consumed?
- How well is it chewed (oral amylase)?
- What type of carbohydrate?
- What is the ratio of carbohydrate to fat, protein, and fiber?
- What is the gastric emptying time?
- What is the pancreatic/brush border digestive enzyme status?
- What is the status of the microbiota?

We are...



- How fast we eat
- How fast we digest
- How fast we absorb
- How fast we assimilate

Gut Microbiome Impact on Glycemic Response



Composition, abundance, and gene expression of the gut microbiota contributes to a highly variable postprandial glucose response (PPGR) among adults.



The mechanism of action is not yet fully known, but the gut microbiota can have indirect effects on metabolism through signaling metabolites – short-chain fatty acids (SCFAs) and bile acids (BAs).

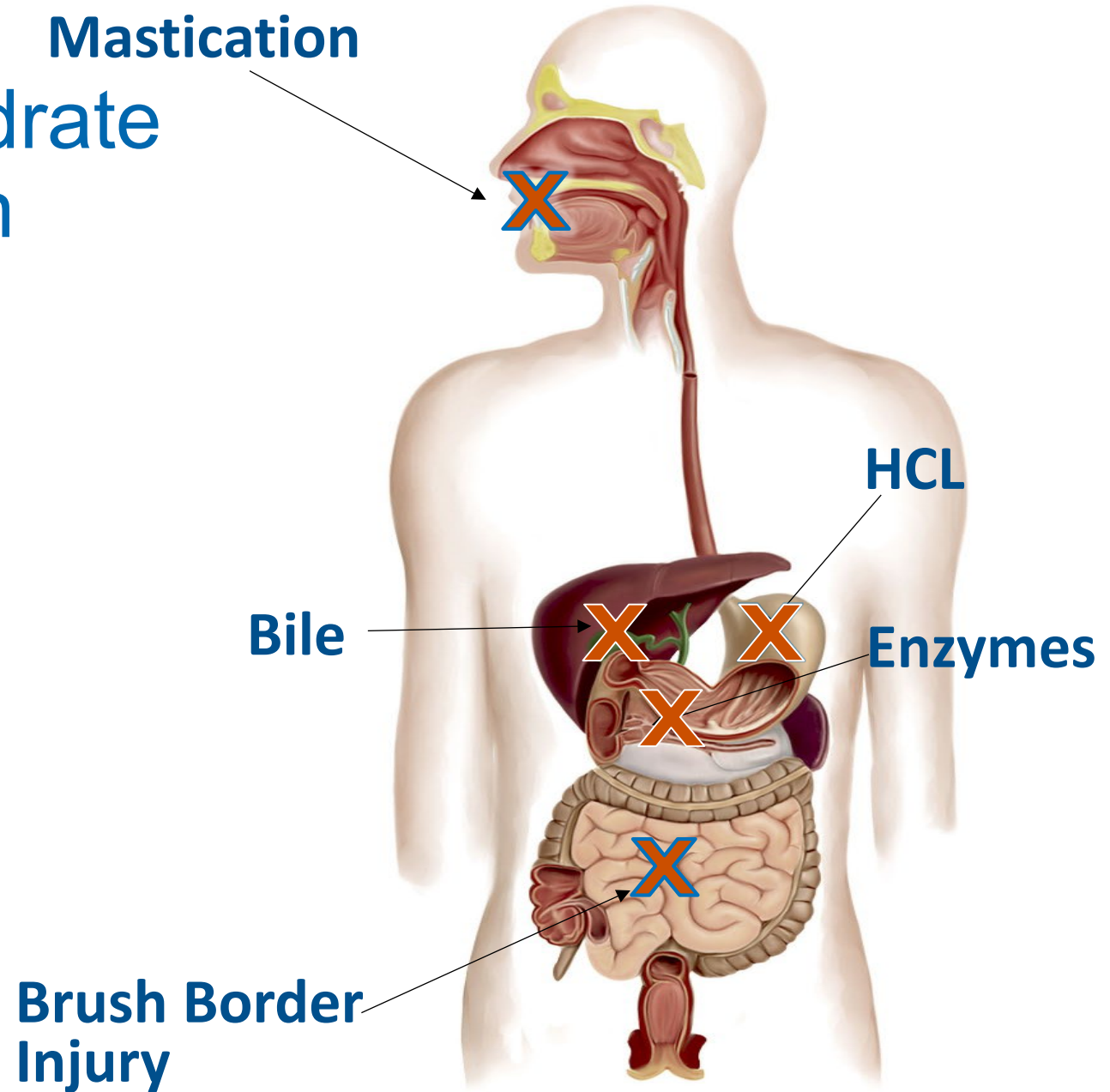


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Impairments in Carbohydrate Digestion and Absorption

- Gastric emptying speed
- Pancreatic insufficiency
- SIBO
- FODMAP intolerance
- Brush border injury
- Dysmotility/diarrhea
- Dysbiosis and GI infections



PFCMVP: Part 2

Macronutrients and Micronutrients

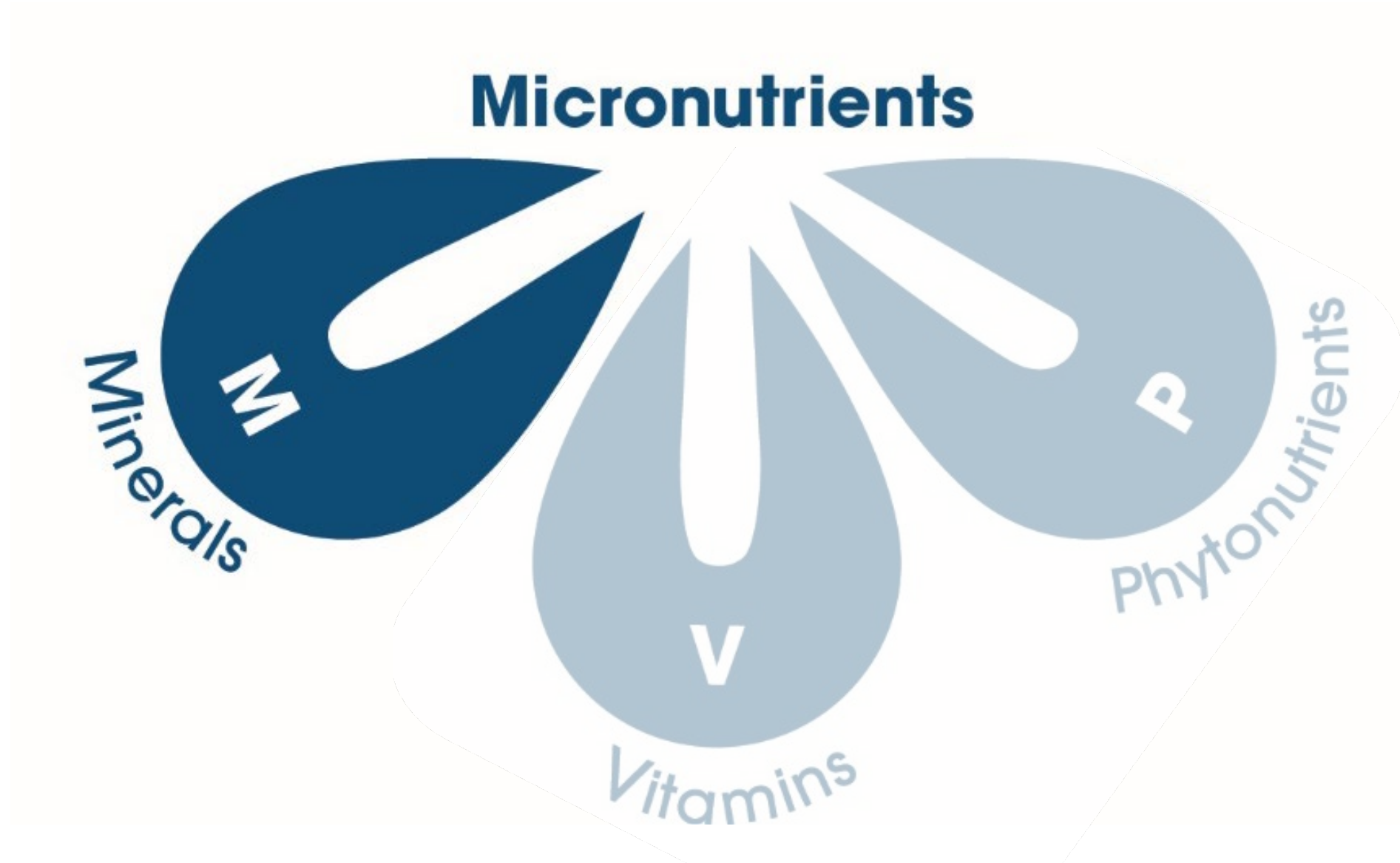
APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- PFCMVP – Minerals



Micronutrients & Phytonutrients



Minerals

Key Features

- 16 inorganic (no carbon) nutrients
- All are “essential” nutrients
- Major minerals
- Trace minerals
- Cofactors and rate-limiting steps of myriad physiological functions

Minerals

EXAMPLES:

Major minerals are minerals that your body needs daily in significant amounts to maintain fluid balances in the body. Individual minerals also have additional health benefits.

Major Mineral	Food Sources
Calcium	Cheese, dark leafy greens (collard greens, kale, mustard greens, spinach, turnip greens, etc.), sardines, sesame seeds, tofu, yogurt
Chloride	Soy sauce, table salt; small amounts in breads, meats, milk, vegetables
Magnesium	Beans (black, navy, soy), cashews, dark leafy greens (spinach, Swiss chard, etc.), quinoa, seeds (pumpkin, sesame, sunflower)
Phosphorus	Fish, eggs, meat, milk, poultry
Potassium	Avocado, banana, beans (lima, soy), beets, dark leafy greens (spinach, Swiss chard), lentils, potato, sweet potato, yogurt
Sodium	Almost all foods contain sodium. Processed foods (especially processed meats) contain extremely high levels and should be limited or avoided. Adding a high-quality salt to your food is a good way to ensure adequate intake.
Sulfur	Beans, eggs, fish, meats, milk, nuts, poultry

Minerals

EXAMPLES:

Trace minerals are minerals that your body only needs a small amount of to function properly. Individual minerals also have additional health benefits.

Trace Mineral	Food Sources
Chromium	Barley, black pepper, broccoli, green beans, lettuce (romaine), oats, tomatoes
Copper	Apricots (dried), asparagus, dark chocolate, dark leafy greens, lentils, liver (beef), mushrooms, nuts, seeds, shrimp
Iodine	Cod, eggs, milk (cow), salmon, sardines, sea vegetables, scallops, shrimp, strawberries, tuna
Iron	Asparagus, beans, bok choy, cumin, dark leafy greens (collard greens, spinach, etc.), eggs, leeks, lentils, organ meats, parsley, poultry, red meats, shellfish, turmeric

Minerals

EXAMPLES:

Additional trace minerals.

Trace Mineral	Food Sources
Manganese	Beans, cinnamon, cloves, dark leafy greens (collard greens, kale, spinach, etc.), oats, pineapple, rice (brown), seeds, turmeric
Molybdenum	Barley, beans, lentils, oats, peas
Selenium	Asparagus, beef, Brazil nuts, clams, cod, mushrooms, organ meats (liver, kidney), oysters, poultry, salmon, sardines, shrimp, tofu, tuna
Zinc	Asparagus, beans, beef, chicken, lamb, mushrooms, oysters, scallops, seeds (pumpkin, sesame), shrimp, spinach

Minerals

Tethered to the Matrix and the Functional Medicine Model

- Key role as cofactors in many physiological processes
- Essential for function in all nodes of the matrix
- Deficiency or insufficiency is a common mediator of dysfunction
- Recall prevalence of mineral deficiencies at population level

Nutrient Declines for 43 Garden Crops

- 43 garden crops were evaluated between 1950 and 1999 for nutrient content using USDA nutrient content data.
- “As a group, **the 43 foods show apparent, statistically reliable declines for 6 nutrients** (protein, Ca, P, Fe, riboflavin, and ascorbic acid).”

Mineral Content Comparison 1930s to 1980s

- A comparison of the mineral content of 20 fruits and 20 vegetables grown in the 1930s and the 1980s shows several **marked reductions in mineral content.**
- The only mineral that showed no significant differences over the 50-year period was phosphorous.
- **“A nutritional problem associated with the quality of food has developed over those 50 years.”**

Toolkit Items – Minerals: Cooking to Preserve Nutrients



Micronutrients are nutrients the body needs small amounts of in order to function properly. This class of nutrients includes minerals, vitamins, and phytonutrients.

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Micronutrients:

Minerals

Minerals are a group of nutrients (most contain carbon) that are essential for human development. The body needs small amounts of these nutrients, because it cannot produce them on its own. They are grouped into two categories: major and trace minerals.

Major Minerals

Major minerals are minerals that your body needs in large amounts in order to function properly. Individual minerals also have additional health benefits.

Major Mineral	Health Benefits
Calcium	• Bone health • Muscle function • Regulation of heart rate
Chloride	• Blood pressure regulation • Electrolyte balance • Regulation of stomach acid
Magnesium	• Bone health • Blood sugar control • Blood pressure regulation • Energy production • Nervous system health
Phosphorus	• Bone health • Cellular energy • DNA and RNA synthesis • Energy production
Potassium	• Blood pressure regulation • Electrolyte balance • Kidney function
Sodium	• Blood pressure regulation • Electrolyte balance • Muscle function
Sulfur	• Cellular energy • Collagen synthesis • Cholesterol regulation • Insulin sensitivity • Metabolism • Skin health



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Trace Minerals

Trace minerals are minerals that your body only needs a small amount of in order to function properly. Individual minerals also have additional health benefits.

Trace Mineral	Health Benefits	Best Food Sources
Chromium	• Blood sugar control • Bone health • Brain health • Digestion • Energy production • Eye health and vision • Heart health • Skin health	Barley, black pepper, broccoli, green beans, lettuce (romaine), oats, tomatoes
Copper	• Antioxidant protection • Bone health • Cell/tissue growth and repair • Cholesterol balance • Energy production • Nervous system health	Apricots (dried), asparagus, dark chocolate, dark leafy greens, lentils, liver (beef), mushrooms, nuts, seeds, shrimp
Fluoride	• Bone and teeth health	Drinking water, fish, tea
Iodine	• Cellular energy • Metabolism • Thyroid function • Thyroid hormone production	Cod, eggs, milk (cow), salmon, sardines, sea vegetables, scallops, shrimp, strawberries, tuna
Iron	• Energy production • Immune system function • Oxygen transport • Red blood cell production • Respiration	Asparagus, beans, bok choy, cumin, dark leafy greens (collard greens, spinach, etc.), eggs, leeks, lentils, organ meats, parsley, poultry, red meats, shellfish, turmeric
Manganese	• Antioxidant protection • Blood sugar control • Bone production and health • Skin, hair, and nail health	Beans, cinnamon, cloves, dark leafy greens (collard greens, kale, spinach, etc.), oats, pineapple, rice (brown), seeds, turmeric
Molybdenum	• Blood plasma health • Enzyme production • Metabolism	Barley, beans, lentils, oats, peas
Selenium	• Antioxidant protection • DNA, protein synthesis • Heart health • Immune system function • Reproductive health • Thyroid function	Asparagus, beef, Brazil nuts, clams, cod, mushrooms, organ meats (liver, kidney), oysters, poultry, salmon, sardines, shrimp, tofu, tuna
Zinc	• Immune system function • Male reproductive health • Metabolism • Sensory organ health (taste, vision) • Skin health	Asparagus, beans, beef, chicken, lamb, mushrooms, oysters, scallops, seeds (pumpkin, sesame), shrimp, spinach

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Version 2



Cooking to Preserve Nutrients

Cooking food offers multiple benefits. It improves food safety, eases digestion, and boosts flavor.¹ However, cooking also affects nutrients in food. Changes to the nutrients in foods are complex and depend on the cooking method and the nutrient type. No cooking method keeps all nutrients perfectly. Still, you can follow some best practices to keep as many nutrients as possible in your food.

QUICK LIST OF COOKING BEST PRACTICES:

- Choose steaming, baking, sous vide, sautéing, and microwaving to preserve nutrients in food.
- Limit grilling, smoking, and frying.
- Bake or steam seafood.
- Steam or microwave cruciferous vegetables.
- Choose fresh fruit when it is in season and as your budget allows.
- Use fats or oils with the right smoke point for your cooking method. See IFM's A Guide to Cooking with Fats and Oils for details.
- Pair vegetables with healthy fats.

Humans have been cooking their food for a long time—approximately 780,000 years.² We cook food to:

- Destroy bacteria, viruses, and parasites that could make us sick³
- Break down high-fiber foods, which may improve our ability to digest nutrients from them^{4,5}
- Make animal proteins easier to digest⁷
- Improve the flavor and texture of food

General Guidelines for Preserving Nutrients

Several factors impact how well foods retain nutrients during cooking. Shorter cooking times, lower cooking temperatures, and using less cooking water (or consuming it, such as in soup or stew) can help you get more nutrients from food. In addition, using healthy cooking oils supports absorption of fat-soluble nutrients like vitamins A and E and carotenoids.^{8,9} For cooking methods that do not use a cooking fat, such as steaming, serve or finish the food with a fat or oil (e.g., avocado, olive oil, nuts, etc.) to help your body absorb fat-soluble vitamins.



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Version 4, updated 2024

Accessible versions in your toolkit.

Toolkit Items – Food Sources (Minerals)



Food Sources: Calcium

Calcium is a mineral that is the primary component of bone. It also participates in communication between cells, supports blood clotting, and helps regulate body temperature. A steady calcium level is essential for bone health.

When there is not enough calcium in the diet, the body uses calcium from bones to maintain a steady blood level and support other functions. Over time, calcium depletion in the bones can lead to osteoporosis, a condition called osteoporosis, which is a loss of bone mass and strength.

Calcium is widely available in many foods, including dairy products, certain vegetables, and fortified foods. Your calcium needs can be met by eating a variety of these foods.

The Recommended Dietary Allowances (RDAs) for calcium are as follows:

- Females, ages 19-50: 1,000 milligrams (mg) per day
- Females, ages 51-70+: 1,200 mg per day
- Males, ages 19-70: 1,000 mg per day
- Males, ages 71+: 1,200 mg per day

Food, Standard Serving Size	Average Calcium Content (in milligrams)
Plain yogurt, 1 cup	415
Canned sardines, 3 ounces	325
Cheddar cheese, 1 ounce	204
Milk, reduced fat, 1 cup	299
Firm tofu, 1/2 cup	854
Collard greens, 1 cup	268
Canned salmon, 3 ounces	188
Sesame seeds, 1/4 cup	884
Kale, 1 cup	93
Broccoli, 1 cup	61
White beans, 1/2 cup	126
Bok choy, 1 cup	91
Dried figs, 1/2 cup	127
Orange, 1 medium	64
Amaranth, 1/2 cup	187

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2. U.S. Department of Agriculture, Agricultural Research Service. National Nutrient Database for Standard Reference. Legacy Version. <https://ndb.nsl.gov/ndb/>. Accessed January 14, 2019.
3. Oregon State University. *Linea Publishing Institute, Micronutrient Information Center: Magnesium*. <http://lpi.oregonstate.edu/inic/magnesium>. Updated November 2018. Accessed August 12, 2020.



Food Sources: Iron

Iron is a mineral found naturally in a variety of foods and is essential for the production of hemoglobin, a protein in red blood cells that carries oxygen throughout the body. Iron is also involved in energy production and DNA synthesis.

Iron plays an important role in hundreds of processes in the body, including oxygen transport and energy production.

The Recommended Dietary Allowances (RDAs) for iron are as follows:

- Adults, ages 19+: 8 milligrams (mg) per day
- Pregnant women: 27 mg per day
- Lactating women: 9 mg per day

Food, Standard Serving Size	Average Iron Content (in milligrams)
Pumpkin seed kernels (shelled, roasted), 1/2 cup	162
Amaranth grain (cooked), 1 cup	160
Brazil nuts (dried, unblanched), 1 ounce (about 6 nuts)	107
Almonds (dry roasted), 1 ounce (about 23 nuts)	90
Spinach (boiled), 1/2 cup	78
Swiss chard (boiled), 1/2 cup	75
Cashews (dry roasted), 1 ounce (about 16 nuts)	74
Peanuts (oil roasted), 1/2 cup	63
Shredded wheat cereal, 2 large biscuits	61
Black beans (cooked), 1/2 cup	60
Edamame (shelled, cooked), 1/2 cup	50
Molasses, 1 tablespoon	48
Hazelnuts (filberts), 1 ounce (about 21 nuts)	46
Avocado (cubed), 1 cup	44
Brown rice (cooked), 1/2 cup	43

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1. U.S. Department of Health and Human Services, National Institutes of Health, Office of Dietary Supplements. Magnesium. <https://ods.od.nih.gov/factsheets/MagnesiumHealthProfessional/>. Updated March 24, 2020. Accessed August 12, 2020.
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Food Sources: Sodium

Sodium is an essential electrolyte that helps control fluid balance in the body. It also helps regulate blood pressure and is involved in nerve function and muscle contraction.

The bioavailability of sodium, or the amount retained and used by the body, may be lower in grains and plant proteins because these foods also contain phytates, which bind to zinc and inhibit its absorption.

The Recommended Dietary Allowances (RDAs) for sodium are as follows:

- Adults, ages 19+: 2,300 milligrams (mg) per day
- Children, ages 1-13: 1,200 mg per day
- Children, ages 14-18: 1,500 mg per day

Food, Standard Serving Size	Average Sodium Content (in milligrams)
Eastern oysters (breaded, fried), 3 ounces/6 medium	74
Eastern oysters (raw), 3 ounces/6 medium	33
Game meat, bison/deer (braised), 3 ounces	7.3
Beef chuck roast (braised), 3 ounces	7.0
Alaska king crab (cooked), 3 ounces	6.5
Ground beef (cooked), 3 ounces	5.3
Lamb shoulder (braised), 3 ounces	5.2
Pork shoulder (braised), 3 ounces	4.1
Breakfast cereal (fortified), 1/2 cup	3.8
Lobster (cooked), 3 ounces	3.4
Baked beans (canned), 1/2 cup	2.9
Pork loin (braised), 3 ounces	2.9
Chicken, dark meat (roasted), 3 ounces	2.4
Pumpkin seeds (dried), 1 ounce/-17 nuts	2.2
Cashews (dry roasted), 1 ounce/-17 nuts	1.8

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3. Oregon State University. *Linea Publishing Institute, Micronutrient Information Center: Zinc*. <http://lpi.oregonstate.edu/inic/zinc>. Updated June 11, 2018. Accessed March 29, 2019.



Food Sources: Iodine

Iodine is a trace element required for the production of thyroid hormones. Because the iodine content of farmed fish varies considerably, it is difficult to determine the precise iodine content of foods.

The highest iodine concentration occurs in the soil of coastal areas. Because of this, seaweed and seafood are the highest natural sources of this important micronutrient. Dairy products are a good dietary source of iodine because of iodine-containing disinfectants used on milking equipment.

The Recommended Dietary Allowances (RDAs) for iodine are as follows:

- Adults, ages 19+: 150 micrograms (mcg) per day
- Pregnant women: 220 mcg per day
- Lactating women: 290 mcg per day

Food, Standard Serving Size	Average Iodine Content (in micrograms)
16-2984*	158
	116
	93
	85
	76
	26
	21
	14
	13

*depending on the type and location to be lower in iodine while kombu

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Food Sources: Magnesium

Magnesium is a mineral found naturally in food and is also available as a supplement. Good magnesium sources include nuts, seeds, legumes, whole grains, and green leafy vegetables. Smaller amounts of the mineral are in some meats, fish, and dairy products.

Magnesium is used widely throughout the body. It supports energy production, blood pressure regulation, blood sugar metabolism, bone health, mood, digestive regularity, and much more. Magnesium deficiency is very common, with almost half of adults in the United States getting less than the RDA. Ask your functional medicine provider for guidance on the form and dose of supplemental magnesium that is best for your health goals.

The Recommended Dietary Allowances (RDAs) for magnesium are as follows:

- Females (including lactating), ages 19-30: 310 milligrams (mg) per day
- Females (including lactating), ages 31+: 320 mg per day
- Females, pregnant, ages 19-30: 350 mg per day
- Females, pregnant, ages 31-50: 360 mg per day
- Males, ages 19-30: 400 mg per day
- Males, ages 31+: 420 mg per day

Food, Standard Serving Size	Average Magnesium Content (in milligrams)
Pumpkin seed kernels (shelled, roasted), 1/2 cup	162
Amaranth grain (cooked), 1 cup	160
Brazil nuts (dried, unblanched), 1 ounce (about 6 nuts)	107
Almonds (dry roasted), 1 ounce (about 23 nuts)	90
Spinach (boiled), 1/2 cup	78
Swiss chard (boiled), 1/2 cup	75
Cashews (dry roasted), 1 ounce (about 16 nuts)	74
Peanuts (oil roasted), 1/2 cup	63
Shredded wheat cereal, 2 large biscuits	61
Black beans (cooked), 1/2 cup	60
Edamame (shelled, cooked), 1/2 cup	50
Molasses, 1 tablespoon	48
Hazelnuts (filberts), 1 ounce (about 21 nuts)	46
Avocado (cubed), 1 cup	44
Brown rice (cooked), 1/2 cup	43

REFERENCES

1. U.S. Department of Health and Human Services, National Institutes of Health, Office of Dietary Supplements. Magnesium. <https://ods.od.nih.gov/factsheets/MagnesiumHealthProfessional/>. Updated March 24, 2020. Accessed August 12, 2020.
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3. Oregon State University. *Linea Publishing Institute, Micronutrient Information Center: Magnesium*. <http://lpi.oregonstate.edu/inic/magnesium>. Updated November 2018. Accessed August 12, 2020.



Food Sources: Selenium

Selenium is an essential trace mineral important for thyroid health, DNA production, reproductive health, immune support, and antioxidant defense. The nutrient is found in both animal and plant-based foods.

Selenium is present in seafood, meats, eggs, some dairy products, grains, nuts, seeds, and other plant foods. Plant sources of the mineral may have variable selenium content, as their uptake of this nutrient depends on the selenium concentration of the soil where they are grown.

The Recommended Dietary Allowances (RDAs) for selenium are as follows:

- Adults, ages 19+: 55 micrograms (mcg) per day
- Pregnant women: 60 mcg per day
- Lactating women: 70 mcg per day

Food, Standard Serving Size*	Average Selenium Content (in micrograms)
Brazil nuts, 1 ounce (6-8 nuts)	544
Tuna (yellowfin), 3 ounces	92
Oysters (Pacific), 3 ounces	66
Clams, 3 ounces	54
Halibut, 3 ounces	47
Sardines (canned), 3 ounces	45
Shrimp, 3 ounces	42
Salmon (wild-caught), 3 ounces	31
Cod (Atlantic), 3 ounces	28
Beef chuck roast, 3 ounces	23
Cottage cheese, 1 cup	20
Sunflower seed butter, 1 tablespoon	17
Egg, whole, hardboiled	15
Oatmeal, cooked with water, 1 cup	13

*Serving size and nutrient amounts for all meat and fish are for the cooked foods.

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3. U.S. Department of Agriculture, Agricultural Research Service. National Nutrient Database (NND) National Nutrient Database for Standard Reference. Legacy Version. <https://ndb.nsl.gov/ndb/>. Accessed January 14, 2019.



Food Sources: Zinc

Zinc is an essential trace mineral; it must be obtained through the diet and is required by the body in small amounts. Zinc is found in shellfish, beef and other red meats, as well as nuts, legumes and fortified breakfast cereals. The bioavailability of zinc, or the amount retained and used by the body, may be lower in grains and plant proteins because these foods also contain phytates, which bind to zinc and inhibit its absorption.

Zinc plays an important role in several body systems – over 100 enzymes in the body require zinc to function. Zinc is necessary for normal growth and development, immune and neurological function, proper sense of taste and smell, and maintaining the structure of cell membranes, as well as protecting them from oxidative damage.

The Recommended Dietary Allowances (RDAs) for zinc are as follows:

- Adults, ages 19+: 8 milligrams (mg) per day
- Pregnant women, ages 19+: 11 mg per day
- Lactating women, ages 19+: 12 mg per day

Food, Standard Serving Size	Average Zinc Content (in milligrams)
Eastern oysters (breaded, fried), 3 ounces/6 medium	74
Eastern oysters (raw), 3 ounces/6 medium	33
Game meat, bison/deer (braised), 3 ounces	7.3
Beef chuck roast (braised), 3 ounces	7.0
Alaska king crab (cooked), 3 ounces	6.5
Ground beef (cooked), 3 ounces	5.3
Lamb shoulder (braised), 3 ounces	5.2
Pork shoulder (braised), 3 ounces	4.1
Breakfast cereal (fortified), 1/2 cup	3.8
Lobster (cooked), 3 ounces	3.4
Baked beans (canned), 1/2 cup	2.9
Pork loin (braised), 3 ounces	2.9
Chicken, dark meat (roasted), 3 ounces	2.4
Pumpkin seeds (dried), 1 ounce/-17 nuts	2.2
Cashews (dry roasted), 1 ounce/-17 nuts	1.8

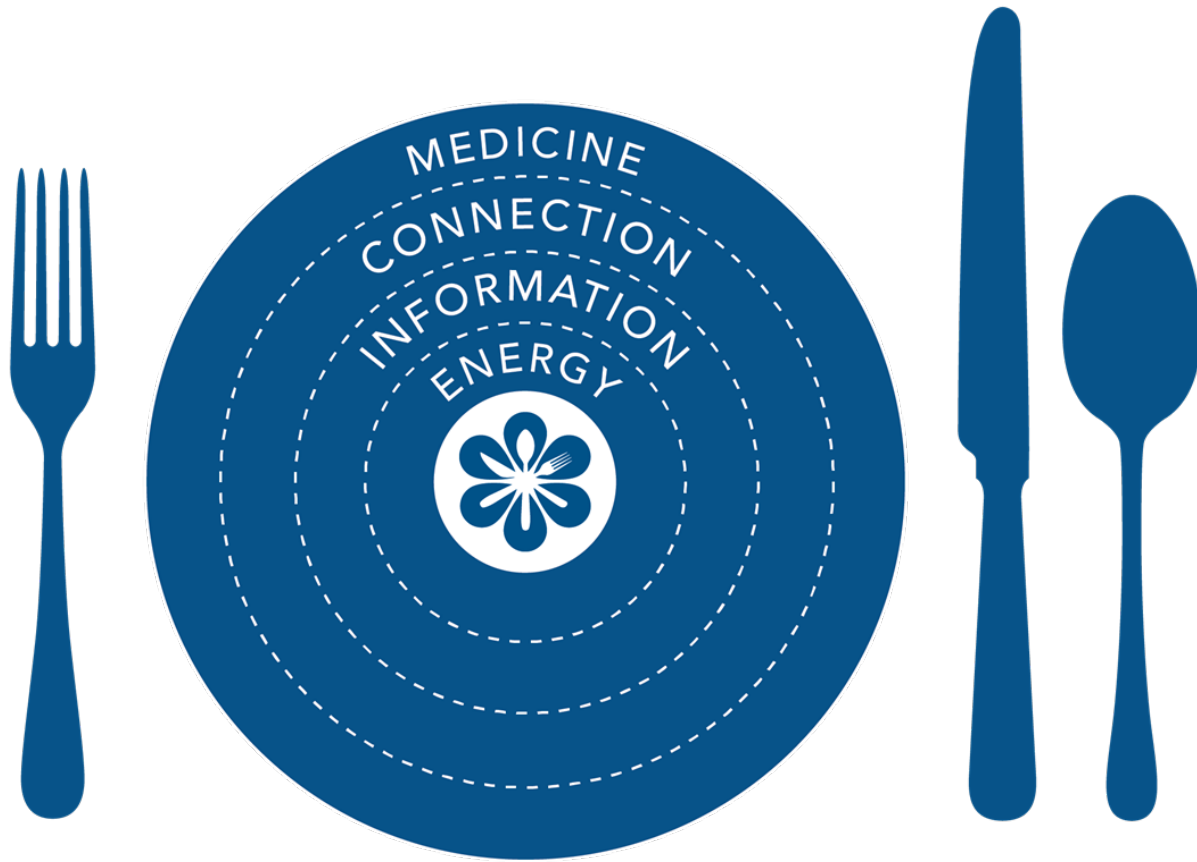
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3. Oregon State University. *Linea Publishing Institute, Micronutrient Information Center: Zinc*. <http://lpi.oregonstate.edu/inic/zinc>. Updated June 11, 2018. Accessed March 29, 2019.

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We are...



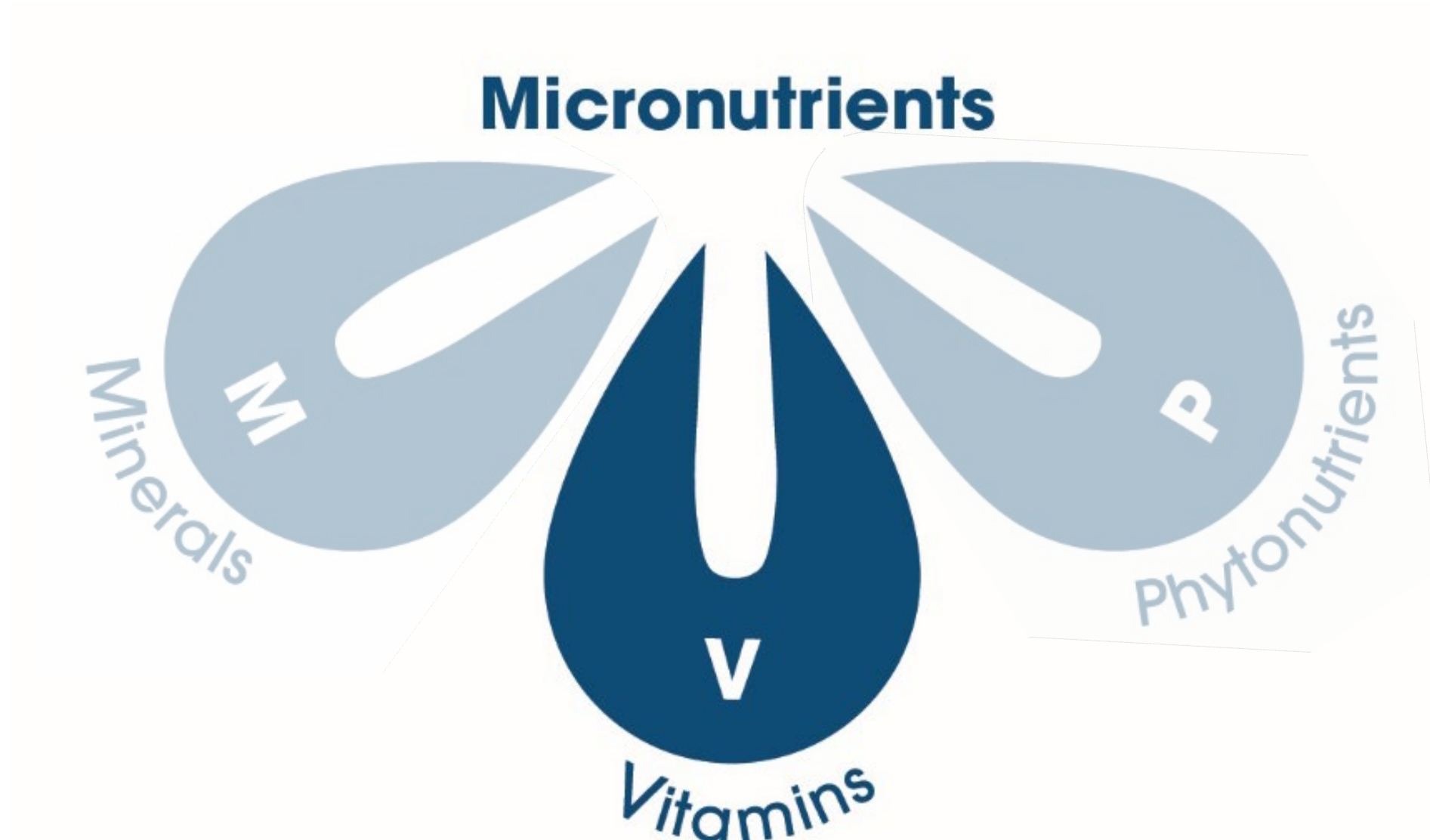
- What we eat ate
- What's in the soil in which what we eat grew
- How we store food
- How we cook food
- How we absorb nutrients
- How we react to food

Clinical Focus

- PFCMVP – Vitamins



Micronutrients & Phytonutrients



Vitamins

Key Features

- 13 are “essential” (not produced by humans)
- Others produced by humans/microbiota, but not adequate amounts
- Water-soluble
- Fat-soluble
- Cofactors and rate-limiting steps of myriad physiological functions

Vitamins

FAT-SOLUBLE:

Fat-Soluble Vitamins	Food Sources
Vitamin A (carotenoids, retinol)	Bell peppers, bok choy, cantaloupe, carrots, dark leafy greens (collard greens, kale, spinach, Swiss chard, etc.), fish, milk, liver, sweet potato, tropical fruits
Vitamin D* (D2, D3)	Egg yolks, fortified foods (milk, orange juice, soy milk, etc.), liver (beef), mackerel, salmon, sardines, tuna *Note: The best source for vitamin D is natural sunlight.
Vitamin E	Almonds, asparagus, avocado, dark leafy greens (beet greens, mustard greens, spinach, Swiss chard, turnip greens, etc.) sunflower seeds
Vitamin K	Asparagus, broccoli, bok choy, Brussels sprouts, cabbage, cauliflower, dark leafy greens (beet greens, collard greens, kale, mustard greens, spinach, Swiss chard, turnip greens, etc.), parsley

Vitamins

WATER-SOLUBLE:

Water-Soluble Food Sources Vitamins	
Biotin (B7)	Almonds, carrots, eggs, liver, oats, onions, peanuts, salmon, sweet potato, tomatoes, whole grain cereals, walnuts
Pantothenic Acid (B5)	Avocado, broccoli, chicken, fish, lentils, mushrooms (cremini, shiitake), organ meats (liver, kidney), peas, shellfish, sweet potato, turkey, yogurt; found in most other foods
Vitamin B1 (Thiamin)	Barley, beans (black, lima, navy, pinto, etc.), lentils, oats, peas, sunflower seeds
Vitamin B2 (Riboflavin)	Almonds, asparagus, beef, broccoli, cheese, chicken, dark leafy greens (beet greens, spinach, etc.), eggs, halibut, milk, mushrooms (cremini), salmon, soybeans, spinach, yogurt

Vitamins

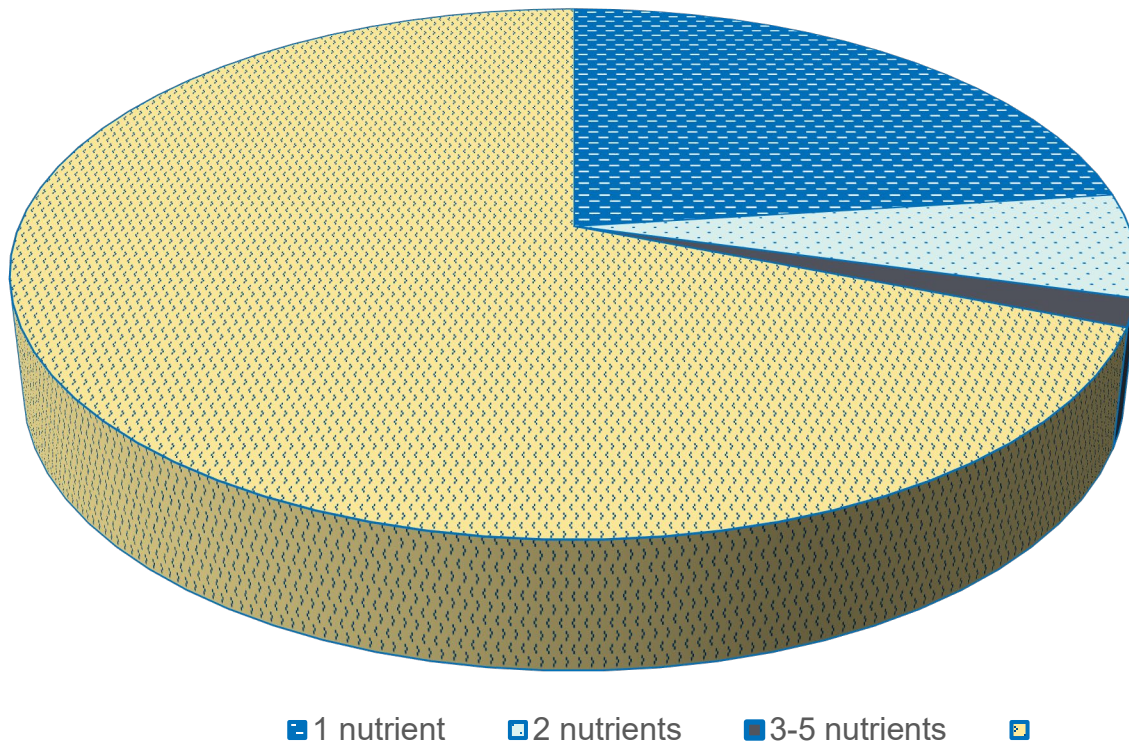
WATER-SOLUBLE:

Water-Soluble Food Sources Vitamins	
Vitamin B3 (Niacin)	Beef, chicken, lamb, rice (brown), salmon, sardines, shrimp, tuna, turkey
Vitamin B6 (Pyridoxine)	Banana, beef, chicken, potatoes, salmon, spinach, sunflower seeds, sweet potato, tuna, turkey
Vitamin B9 (Folate)	Asparagus, beans (black, garbanzo, kidney, navy, pinto, etc.), broccoli, dark leafy greens (spinach, turnip greens, etc.), lentils
Vitamin B12 (Cobalamin)	Beef, cheese, chicken, clams, cod, crab, eggs, lamb, mackerel, milk (cow), mussels, rockfish, salmon, sardines, scallops, shrimp, tuna, turkey, yogurt
Vitamin C	Bell peppers, broccoli, Brussels sprouts, cabbage, cantaloupe, cauliflower, dark leafy greens, grapefruit, kiwi, lemons, limes, orange, papaya, pineapple, potato, spinach, strawberries, sweet red pepper, tomato

NHANES USA – RISK OF DEFICIENCY IN MICRONUTRIENTS

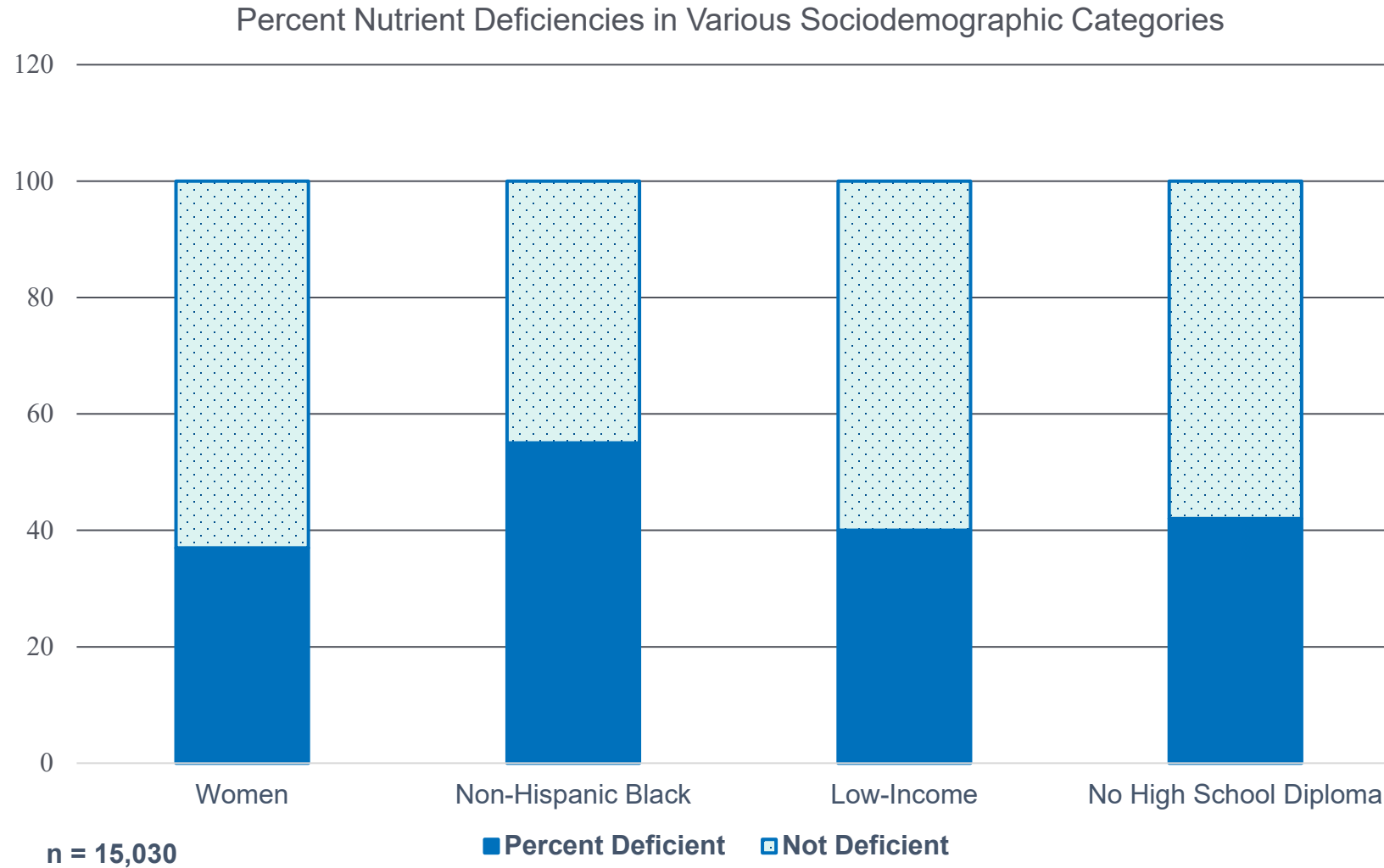
(n>15,000)

% Population With Nutrient Deficiency

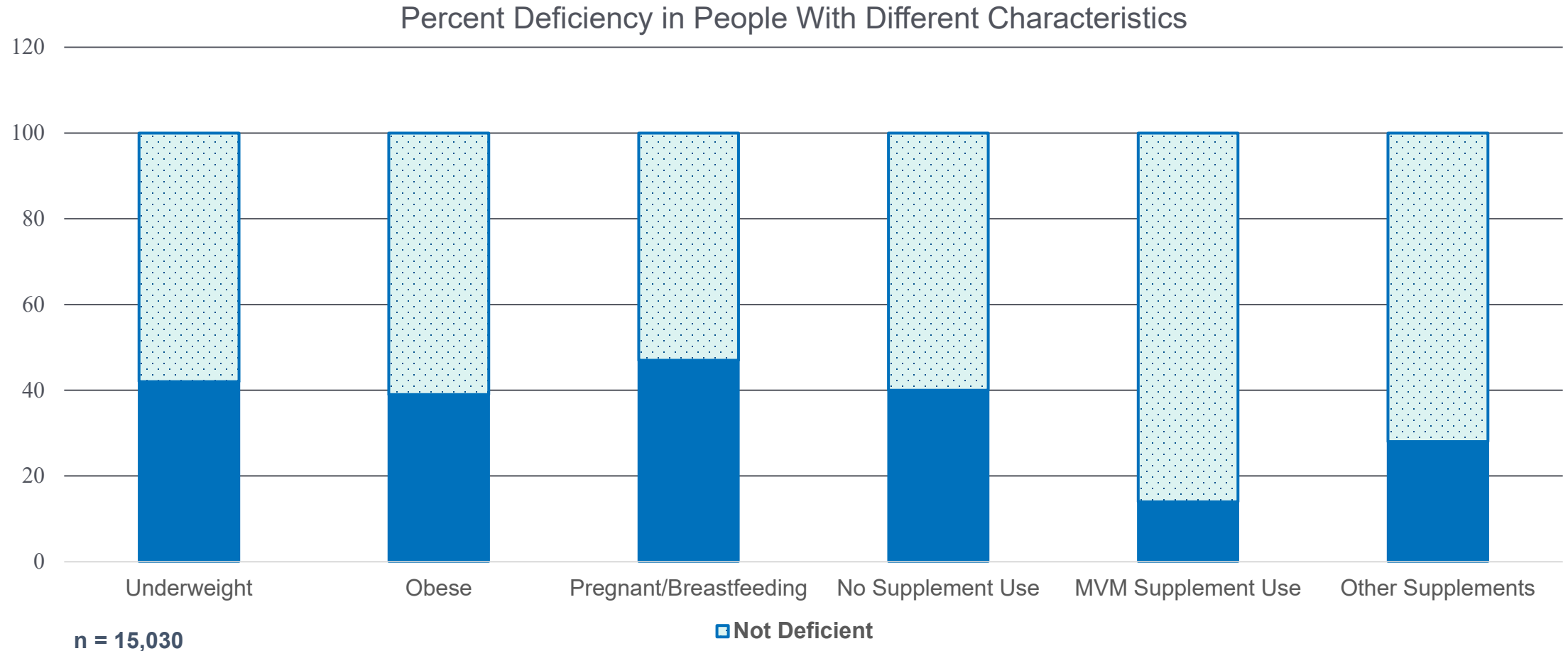


- 23% have 1 nutrient deficiency
- 6.3% have 2 nutrient deficiencies
- 1.7% have 3-5 nutrient deficiencies

NHANES USA – Risk of Deficiency in Micronutrients



NHANES USA – Risk of Deficiency in Micronutrients



Toolkit Items – Vitamins: Cooking to Preserve Nutrients



Micronutrients are nutrients the body needs small amounts of in order to function properly. This class of nutrients includes minerals, vitamins, and phytonutrients.

Micronutrients:

Vitamins

Vitamins are a group of nutrients that are essential for normal cell function, growth, and development. While some vitamins are made by the body, there are 13 vitamins that the human body can't manufacture. They are known as essential vitamins, because it is essential for us to consume them. Essential vitamins are grouped into two categories: fat-soluble and water-soluble.

Fat-Soluble Vitamins

Vitamins that require fat and bile for absorption in the body are known as fat-soluble vitamins. They are stored in the fat cells and liver until the body needs them, so they do not need to be consumed every day. The body can't eliminate these vitamins as easily as water-soluble vitamins. In fact, excess fat-soluble vitamins can accumulate in the body and cause damage, so it's important to consume them in appropriate amounts. There are four essential fat-soluble vitamins: Vitamins A, D, E, and K. The table below lists these vitamins with their best food sources.

Fat-Soluble Vitamin	Health Benefits	Best Food Sources
Vitamin A	• Cell growth and development • Controlling inflammation • Eye health and vision • Gene expression • Immune system function • Prenatal and postnatal development • Red blood cell production • Reproductive health	• Bell peppers, bok choy, cantaloupe, carrots, dark leafy greens (collard greens, kale, spinach, Swiss chard, etc.), fish, milk, liver, sweet potato, tropical fruits
Vitamin D*	• Blood sugar regulation • Bone development and maintenance • Cell growth and development • Immune system function • Insulin secretion • Metabolism	• Egg yolks, fortified foods (milk, orange juice, soy milk, etc.), liver (beef), mackerel, salmon, sardines, tuna *Note: The best source for vitamin D is natural sunlight.
Vitamin E	• Antioxidant protection • Cognition and brain health • Eye health and vision • Immune system function	• Almonds, asparagus, avocado, dark leafy greens (beet greens, mustard greens, spinach, Swiss chard, turnip greens, etc.), sunflower seeds
Vitamin K	• Blood clotting • Bone development and maintenance • Cardiovascular health • Cellular function • Soft tissue integrity (prevents calcification)	• Asparagus, broccoli, bok choy, Brussels sprouts, cabbage, cauliflower, dark leafy greens (beet greens, collard greens, kale, mustard greens, spinach, Swiss chard, turnip greens, etc.), parsley

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Water-soluble vitamins are known as water-soluble vitamins. They are not stored in tissues. Essential water-soluble vitamins are any water-soluble vitamins that the body doesn't store in urine.

Health Benefits	Best Food Sources
• Energy balance • Bone development • Eye and nail health	• Almonds, carrots, eggs, liver, oats, onions, peanuts, salmon, sweet potato, tomatoes, whole-grain cereals, walnuts
• Immune health • Blood production • Bone health • Wound healing	• Avocado, broccoli, chicken, fish, lentils, mushrooms (crimini, shiitake), organ meats (liver, kidney), peas, shellfish, sweet potato, turkey, yogurt; found in most other foods
• Cancer prevention/cancer treatment • Immune health • Brain health • Blood production • Immune system support	• Barley, beans (black, lima, navy, pinto, etc.), lentils, oats, peas, sunflower seeds
• Cancer prevention/cancer treatment • Immune health • Blood production • Immune system support	• Almonds, asparagus, beef, broccoli, cheese, chicken, dark leafy greens (beet greens, spinach, etc.), eggs, halibut, milk, mushrooms (crimini), salmon, soybeans, spinach, yogurt
• Cancer prevention/cancer treatment • Immune health • Blood production • Immune system support	• Beef, chicken, lamb, rice (brown), salmon, sardines, shrimp, tuna, turkey
• Nervous system support • Immune health • Blood production • Immune system support	• Banana, beef, chicken, potatoes, salmon, spinach, sunflower seeds, sweet potato, tuna, turkey
• Nervous system support • Immune health • Blood production • Immune system support	• Asparagus, beans (black, garbanzo, kidney, navy, pinto, etc.), broccoli, dark leafy greens (spinach, turnip greens, etc.), lentils

• Nervous system support
• Immune health
• Blood production
• Immune system support

• Beef, cheese, chicken, clams, cod, crab, eggs, lamb, mackerel, milk (cow), mussels, rockfish, salmon, sardines, scallops, shrimp, tuna, turkey, yogurt

• Bell peppers, broccoli, Brussels sprouts, cabbage, cantaloupe, cauliflower, dark leafy greens, grapefruit, kiwi, lemons, limes, orange, papaya, pineapple, potato, spinach, strawberries, sweet red pepper, tomato

Source: <http://fp.oregonstate.edu/mic>. Published October 21, 2016. Accessed 10/21/2016.

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Version 2



Cooking to Preserve Nutrients

Cooking food offers multiple benefits. It improves food safety, eases digestion, and boosts flavor.¹ However, cooking also affects nutrients in food. Changes to the nutrients in foods are complex and depend on the cooking method and the nutrient type. No cooking method keeps all nutrients perfectly. Still, you can follow some best practices to keep as many nutrients as possible in your food.

QUICK LIST OF COOKING BEST PRACTICES:

- Choose steaming, baking, sous vide, sautéing, and microwaving to preserve nutrients in food.
- Limit grilling, smoking, and frying.
- Bake or steam seafood.
- Steam or microwave cruciferous vegetables.
- Choose fresh fruit when it is in season and as your budget allows.
- Use fats or oils with the right smoke point for your cooking method. See IFM's *A Guide to Cooking with Fats and Oils* for details.
- Pair vegetables with healthy fats.

Humans have been cooking their food for a long time—approximately 780,000 years.² We cook food to:

- Destroy bacteria, viruses, and parasites that could make us sick³
- Break down high-fiber foods, which may improve our ability to digest nutrients from them^{4,5}
- Make animal proteins easier to digest^{6,7}
- Improve the flavor and texture of food

General Guidelines for Preserving Nutrients

Several factors impact how well foods retain nutrients during cooking. Shorter cooking times, lower cooking temperatures, and using less cooking water (or consuming it, such as in soup or stew) can help you get more nutrients from food. In addition, using healthy cooking oils supports absorption of fat-soluble nutrients like vitamins A and E and carotenoids.^{8,9} For cooking methods that do not use a cooking fat, such as steaming, serve or finish the food with a fat or oil (e.g., avocado, olive oil, nuts, etc.) to help your body absorb fat-soluble vitamins.



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Version 2

Toolkit Items – Food Sources (Vitamins)



Food Sources: Vitamin A

Vitamin A is the term used to describe a group of fat-soluble compounds which are available in the diet in two different forms: preformed vitamin A (retinol) and provitamin A carotenoids, such as beta-carotene. The richest sources of preformed vitamin A include organ meats and fish oils, with smaller amounts found in dairy products and fortified cereals. Provitamin A carotenoids are found predominantly in orange, yellow, and green colored fruits and vegetables.

Vitamin A is essential to support cell growth, immune function, and vision. Deficiency of vitamin A is a major, preventable cause of blindness.

Food Sources: Vitamin C

Vitamin C, or ascorbic acid, is a water-soluble nutrient required for protein metabolism, immune function, wound healing, gene expression, and collagen production.¹ Vitamin C is also an important antioxidant and participates in many chemical reactions in the body.²

Fruits and vegetables contain the most vitamin C of all food sources. Some fortified foods may also contain vitamin C, as well as, supplements. Since vitamin C is heat sensitive, eating fruits and vegetables either raw, or lightly steamed is recommended. Vitamin C deficiency is rare as long as fruits and vegetables are included in the diet. Individuals who smoke or eat few fruits and vegetables may be

Food Sources: Vitamin D

Vitamin D is naturally found in very few foods, including fish liver oils, some fatty fish, beef liver and egg yolks. This fat-soluble vitamin can also be obtained from fortified foods such as orange juice, milk, and ready-to-eat breakfast cereals or dietary supplements.

At least some vitamin D needs are met through skin exposure to sunlight. How much vitamin D is made depends on the time of day, age, how much skin is uncovered, and skin tone. Without sunblock and with arms and legs exposed, the skin will make an average of 10,000 to 15,000 units of vitamin D in one hour of exposure. Sunblock with an SPF of more than 15 blocks 100% of vitamin

Food Sources: Vitamin E

Vitamin E refers to a family of eight, fat-soluble nutrient compounds. One of these, called alpha-tocopherol, is naturally found in food and is preferentially used by the body.

Vitamin E is an antioxidant which also supports the immune system, protects the lining of blood vessels, and positively influences several metabolic processes in the body.^{1,2} Nuts, seeds, vegetable oils, and leafy greens contain vitamin E. Some food products contain vitamin E as a preservative and it can also be consumed via supplements.



Food Sources: Vitamin B2 (Riboflavin)

Vitamin B2, commonly referred to as riboflavin, is a water-soluble essential nutrient. It is necessary for two important enzymes in the body, which are required for energy production and a wide variety of functions within cells.

Riboflavin helps maintain healthy levels of homocysteine, an amino acid that in high levels can contribute to heart and other diseases.¹ Milk, liver, almonds, and spinach are good sources of vitamin B2.² Exposure to light can deactivate riboflavin, which is why milk is commonly stored in opaque plastic or cardboard containers. Vitamin B2 is used as an orange-red food color additive.

Select dietary sources of riboflavin are noted in the table below.^{1,2}

The Recommended Dietary Allowances (RDAs) for vitamin B2 are as follows:

- Females, ages 19+: 1.1 milligrams (mg) per day
- Males, ages 19+: 1.3 mg per day
- Pregnancy: 1.4 mg per day
- Lactation: 1.6 mg per day

Food, standard serving size	Average Riboflavin Content (in milligrams)
Liver (beef, poultry) 3 ounces	3
Breakfast cereal, fortified, 1 serving	1.3
Plain yogurt, 1 cup	0.6
Whole milk, 1 cup	0.4
Beef tenderloin, 3 ounces	0.3
Almonds, 1 ounce	0.3
Salmon, 3 ounces	0.2
Mushrooms, crimini, ½ cup	0.2
Egg, hardboiled, 1 large	0.2
Chicken breast, 3 ounces	0.2
Spinach, raw, 1 cup	0.1
Apple, 1 large	0.1

REFERENCES

1. U.S. Department of Health and Human Services, National Institutes of Health, Office of Dietary Supplements. Riboflavin. <https://ods.od.nih.gov/factsheets/RiboflavinHealthProfessional/>. Updated March 4, 2020. Accessed May 18, 2020.
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Food Sources: Vitamin B6

Vitamin B6 refers to a family of six water-soluble compounds which are naturally present in many foods, and available in dietary supplements. The six compounds commonly referred to as B6 are: Pyridoxine (PN), Pyridoxine 5'-phosphate (P5P), Pyridoxal (PL), Pyridoxal 5'-phosphate (PLP), Pyridoxamine (PM), and Pyridoxamine 5'-phosphate (PMP).

Vitamin B6 is involved in protein metabolism, immune function, neurotransmitter production, blood sugar production, and formation of healthy red blood cells.¹ It also helps regulate homocysteine, an amino acid that in high amounts can contribute to heart and other diseases.¹ Vitamin B6 is found in a wide variety of foods. The richest sources of vitamin B6 include fish, organ meats, potatoes, and starchy vegetables.² Because plant foods contain a form of vitamin B6 which is less available to the body, strict vegans may need to eat fortified foods or discuss supplementation with their functional medicine practitioner.¹

Select dietary sources of vitamin B6 are noted in the table below.^{1,2}

Food, standard serving size	Average Vitamin B6 Content (in milligrams)
Chickpeas, cooked, 1 cup	1
Fortified breakfast cereal, 1 cup	0.5 – 2.5
Beef liver, 3 ounces	1
Potato, Russet, 1 medium	0.7
Salmon, cooked, 3 ounces	0.6
Turkey, 3 ounces	0.4 – 0.7
Chicken breast, 3 ounces	0.5
Banana, 1 medium	0.4
Spinach, raw, 1 cup	0.4
Winter squash, ½ cup	0.2
Cottage cheese, ½ cup	0.1

REFERENCES

1. U.S. Department of Health and Human Services, National Institutes of Health, Office of Dietary Supplements. Vitamin B6. <https://ods.od.nih.gov/factsheets/VitaminB6HealthProfessional/>. Updated February 24, 2020. Accessed May 18, 2020.
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Food Sources: Folate

Folate is a water-soluble B vitamin that occurs naturally in food. Folic acid is the synthetic form of folate that is found in some supplements and most fortified foods. Leafy green vegetables (e.g., spinach and kale), citrus fruits, and legumes are all natural sources of folate.

Folate helps the body produce and maintain new cells. This is especially important during rapid growth and cell division, such as infancy and pregnancy.¹ Folate is needed to make new DNA and helps prevent damage to DNA that may lead to cancer. Adults and children need folate to make normal red blood cells and prevent anemia. Folate is also essential for the metabolism of homocysteine, an amino acid that in high amounts may contribute to heart disease, dementia, and depression.^{1,2}

The RDAs for folate are expressed as dietary folate equivalents (DFEs) to reflect a higher bioavailability of folic acid compared to natural folate from food.² Some individuals, like those with genetic variants in MTHFR, may respond better to natural folate than folic acid.¹ Talk to your functional medicine practitioner if you have questions or concerns about your folate intake.

Food, Standard Serving Size	Average DFE Content (in micrograms)
Organic, grass-fed beef liver (cooked), 3 ounces	215
Lentils (cooked), ½ cup	180
Pinto beans (cooked), ½ cup	145
Garbanzo beans (cooked), ½ cup	140
Asparagus (cooked), 6 spears	134
Spinach (cooked), ½ cup	130
Black-eyed peas (cooked), ½ cup	105
Cereal, fortified to 25% of Daily Value, 1 serving	100
Brussels sprouts (cooked), ½ cup	80
Beets (cooked), ½ cup	68
Romaine lettuce, 1 cup	64
Avocado, ½ cup	60
Broccoli (cooked), ½ cup	50-60
Orange, 1 medium	40

REFERENCES

1. Oregon State University, Linus Pauling Institute, Micronutrient Information Center. Folate. <https://lpi.oregonstate.edu/info/summary/folate/>. Updated June 2014. Accessed March 18, 2020.
2. U.S. Department of Health and Human Services, National Institutes of Health, Office of Dietary Supplements. Folate. <https://ods.od.nih.gov/factsheets/FolateHealthProfessional/>. Updated March 11, 2020. Accessed March 18, 2020.
3. U.S. Department of Agriculture, Agricultural Research Service. FoodData Central. <https://fdc.nal.usda.gov/>. Accessed March 25, 2020.



Food Sources: Vitamin B12

Vitamin B12 is a water-soluble nutrient required to make healthy red blood cells, support brain health, make DNA, metabolize homocysteine, and prevent pernicious anemia.¹ Vitamin B12 typically comes from animal sources of protein (fish, meat, poultry, eggs), fortified foods (cereals, nut "milks"), and supplements, but is not found in fruits or vegetables.

There are several forms of vitamin B12 including cyanocobalamin, methylcobalamin, deoxycobalamin, and hydroxycobalamin. Those who have high homocysteine levels or genetic variants in methylation genes may respond better to "active" forms of vitamin B12 such as methylcobalamin.²

Strict vegetarians and vegans are at higher risk for vitamin B12 deficiency because of the avoidance of animal foods. Those with celiac disease, Crohn's disease, low stomach acid, and the elderly may also be at higher risk for vitamin B12 deficiency because of poor absorption.¹ Certain medications (proton pump inhibitors, H2 receptor antagonists, Metformin) may also increase the likelihood of vitamin B12 deficiency.¹

Allowances as follows:

microgram per day

Food, standard serving size	Average Vitamin B12 Content (in micrograms)
Clams, 3 ounces	84
Grass-fed beef liver, 3 ounces	70
King crab, 3 ounces	9.8
Lean beef, 3 ounces	7
Wild-caught Rainbow trout, 3 ounces	5.4
Sockeye salmon, 3 ounces	4.8
Light canned tuna, 3 ounces	2.5
Nutritional yeast, fortified, 1 serving	2.4
Haddock, 3 ounces	1.8
Cow milk, 1 cup	1.2
Yogurt, 8 ounces	1.1
Swiss cheese, 1 ounce	0.9
Roasted turkey, 3 ounces	0.8
Hard-boiled egg, 1 large	0.6

REFERENCES

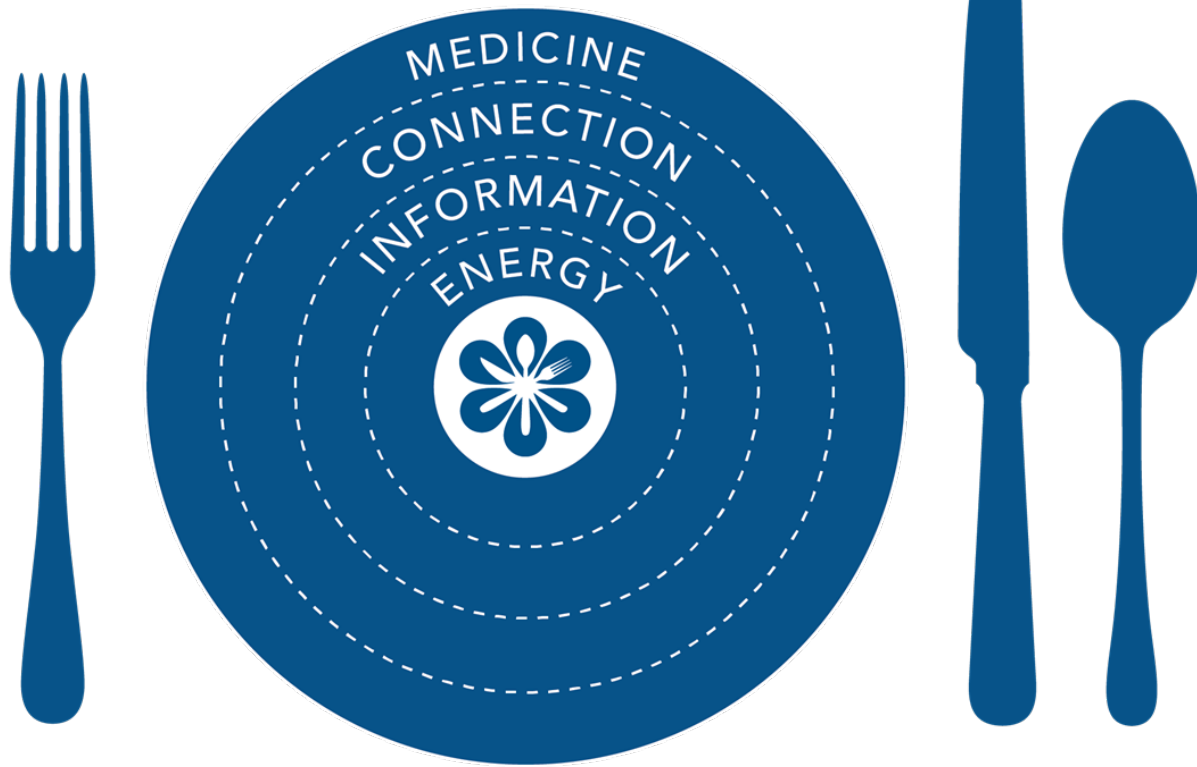
1. U.S. Department of Health and Human Services, National Institutes of Health, Office of Dietary Supplements. Vitamin B12. <https://ods.od.nih.gov/factsheets/VitaminB12HealthProfessional/>. Updated February 19, 2020. Accessed March 25, 2020.
2. Oregon State University, Linus Pauling Institute, Micronutrient Information Center. Vitamin B12. <https://lpi.oregonstate.edu/info/summary/vitamin-b12/>. Updated June 2015. Accessed March 20, 2020.
3. U.S. Department of Agriculture, Agricultural Research Service. FoodData Central. <https://fdc.nal.usda.gov/>. Accessed March 25, 2020.

Vitamins

Tethered to the Matrix and the Functional Medicine Model

- Key role as cofactors in many physiological processes
- Essential for function in all nodes of the matrix
- Deficiency or insufficiency is a common mediator of dysfunction
- Recall prevalence of vitamin deficiencies at population level

We are...



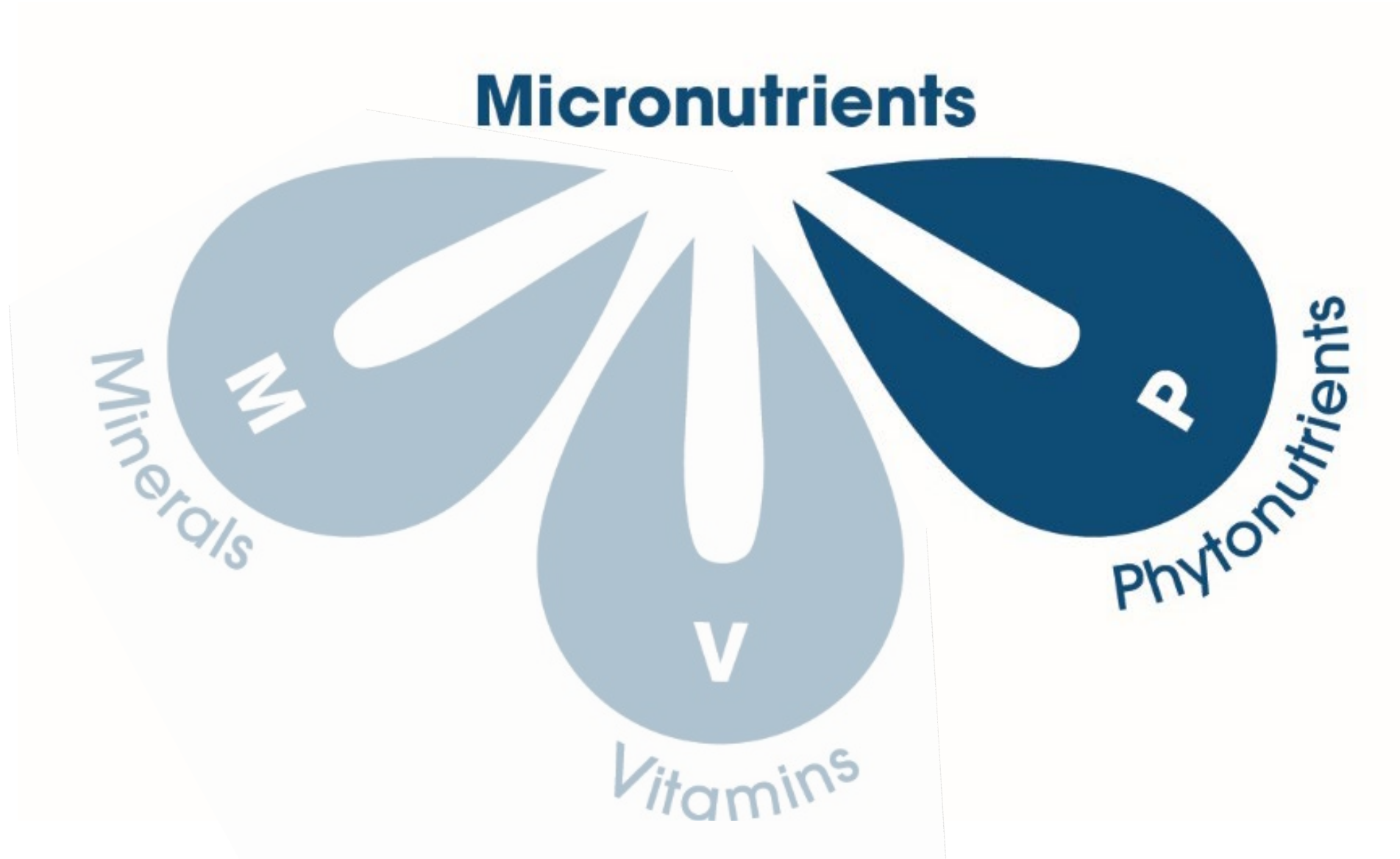
- What we eat ate
- What our microbiota synthesize
- What's in the soil in which what we eat grew
- Where our food grew
- How we store food
- How we cook food

Clinical Focus

- PFCMVP – Phytonutrients



Micronutrients & Phytonutrients



Phytonutrients

Key Features

- Impart color, taste, and smell in all plant foods
- Protect plant from pests and stressors
- Reduced content in non-organic or monoculture systems
- Involved in hormone metabolism, immune system, toxin elimination
- Antioxidant, anti-inflammatory, anti-cancer properties

Phytonutrients

Key Features

- Best preserved in whole, unprocessed foods
- Consumption of a variety of colors daily is optimal
- Cooking and storage techniques may affect quantity
- Deficiencies highly prevalent in consumers of ultra-processed foods
- Found in spices and teas

Phytonutrients – Potential Health Benefits

Red Foods

- Reduced inflammation
- Reduced total and LDL cholesterol
- Reduced blood pressure
- Lower risk of CVD
- Improved memory and cognitive function
- Reduced risk of lung, head, neck, prostate, and breast cancers
- Increased beneficial gut bacteria
- Increased protection of the skin from sun damage

Orange Foods

- Decreased risk of type 2 diabetes
- Lower risk of coronary heart disease (CHD)
- Reduced risk of breast, cervical, head, neck, and bladder cancers
- Lower risk of osteoporosis and bone fractures
- Reduced risk of asthma
- Healthy sperm count

Yellow Foods

- Lower risk of asthma
- Reduced risk of metabolic syndrome and type 2 diabetes
- Decreased risk of stroke
- Healthy vision
- Lower risk of breast, head, neck, bladder, and prostate cancers
- Protection against cell damage

Phytonutrients – Potential Health Benefits

Green Foods

- Reduced risk of lung, bladder, colon, ovarian, pancreatic, stomach, prostate, and kidney cancers
- Lower risk of depression
- Decreased risk of coronary heart disease (CHD) and stroke
- Healthy brain function with aging
- Protection against cell damage
- Enhanced detoxification of harmful chemicals
- Healthy estrogen detoxification

Blue/Purple/Black Foods

- Maintenance of healthy cognitive function
- Decreased blood pressure
- Reduced risk of cardiovascular disease
- Protection from liver damage caused by inflammation
- Reduced fasting blood sugar and hemoglobin A1c
- Less muscle soreness and faster recovery after exercise
- Healthy levels of immune cells that defend the body against harmful microbes
- Protection against DNA damage

White/Tan/Brown Foods

- Decreased risk of breast, colon, esophageal, stomach, and pancreatic cancers
- Decreased total and LDL cholesterol
- Reduced risk of cardiovascular disease; supports heart health by protecting inner walls of blood vessels
- Decreased risk of type 2 diabetes

The Power of Phytochemical Diversity

“These findings indicate that botanical diversity plays a role in determining the bioactivity of high-VF* diets and that **smaller amounts of many phytochemicals may have greater beneficial effects than larger amounts of fewer phytochemicals.**”

*VF = vegetable and fruit



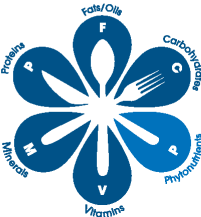
Toolkit – Micronutrients: Phytonutrients



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Micronutrients: Phytonutrients

Phytonutrients, or phytochemicals, are natural compounds found in all plant foods. They are responsible for giving plant foods their distinct colors, tastes, and smells. They also provide plants with protection from pests and environmental stressors. In the human body, phytonutrients play a role in many processes, including (but not limited to) toxin elimination, hormone metabolism, and strengthening the immune system.



Micronutrients are nutrients the body needs small amounts of in order to function properly. This class of nutrients includes minerals, vitamins, and phytonutrients.

Sources of Phytonutrients

More than 25,000 phytonutrients are found in plant foods. Fruits and vegetables are rich sources of phytonutrients, as well as whole grains, legumes, herbs, spices, nuts, seeds, and teas. It is important to eat a variety of colorful phytonutrient-rich fruits and vegetables each day, as phytonutrients and their health benefits are associated with specific colors of foods.

Aim for one to two of each color per day. In order to get the full benefit of phytonutrients, eat these foods in their whole natural form. The list below shows health benefits associated with each color of foods:

- **Red Foods:** anti-bacterial, anti-cancer, anti-inflammatory, blood vessel health, brain health, cell protection, heart health, prostate health
- **Orange Foods:** anti-inflammatory, blood vessel health, brain health, cell protection, heart health, reproductive health
- **Yellow Foods:** anti-inflammatory, cell protection, digestive health, eye health, heart health, immune health
- **Green Foods:** anti-cancer, anti-inflammatory, blood vessel health, bone health, brain health, cell protection, heart health, hormone health, metabolic health
- **Blue/Purple/Black Foods:** anti-inflammatory, blood vessel health, bone health, brain health, cell protection, digestive health, heart health, liver health
- **White/Tan/Brown Foods:** anti-cancer, anti-inflammatory, blood vessel health, bone health, brain health, cell protection, digestive health, heart health, immune health, metabolic health

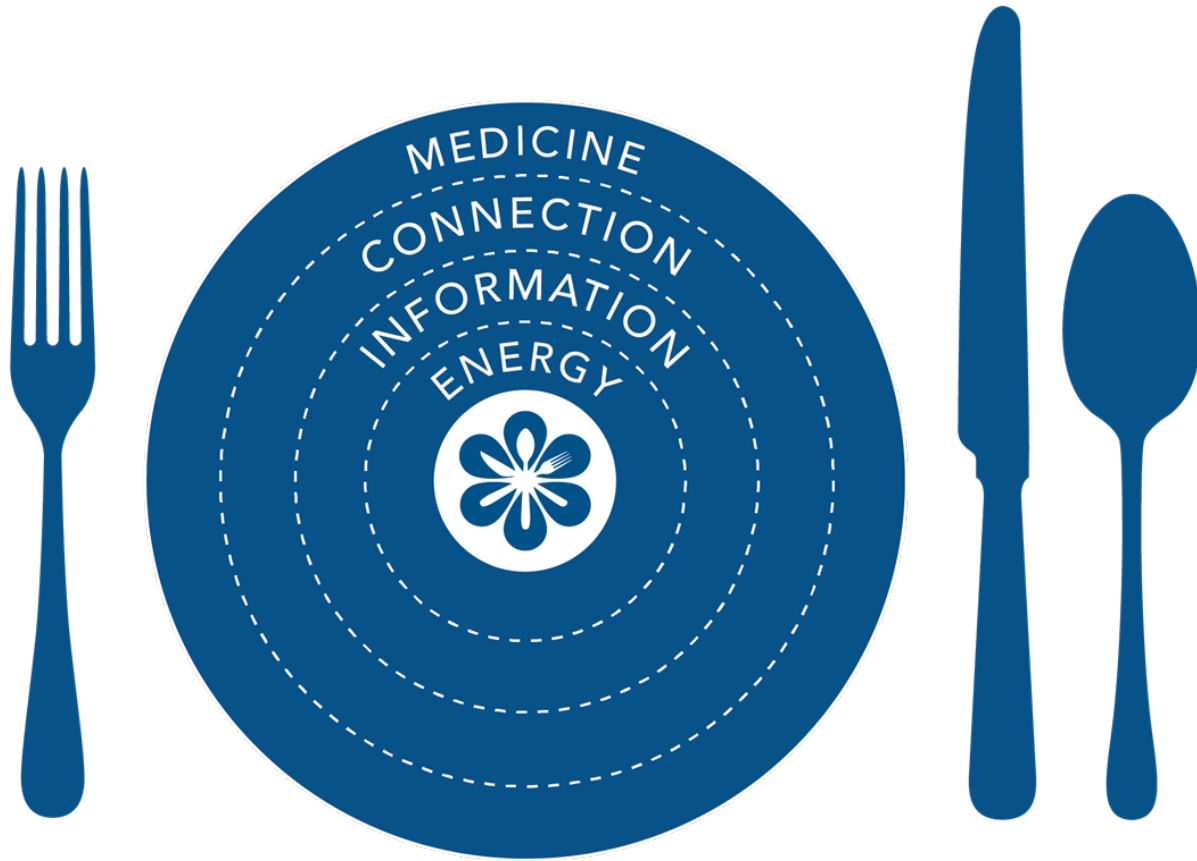
To learn more about phytonutrients and the foods that contain them, ask your functional medicine practitioner about IFM's [Phytonutrient Spectrum](#) suite of documents.



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Version 3

We are...



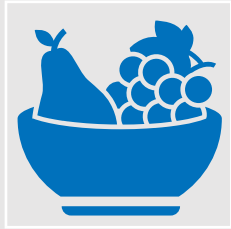
- What colors we eat
- What's in the soil in which what we eat grew
- How we store food
- How we cook food
- How our food is grown

Clinical Focus

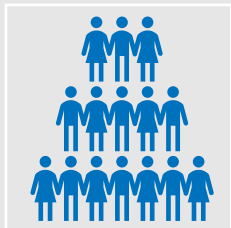
- Summary, conclusion, and key takeaways.



Key Clinical Takeaways



For optimal health, it is important to eat a sufficient quantity and diversity of six nutrient categories on a daily basis:
Protein, Fats and Oils, Carbohydrates, Minerals, Vitamins, and Phytonutrients



Micronutrient deficiencies are highly prevalent in the general population.

Key Clinical Takeaways



Factors related to animal husbandry, soil quality, industrial processing, and food preparation and storage techniques may affect the nutrient density of the foods we consume.



Regardless of the amount of nutrients consumed, factors related to digestion and absorption may affect the assimilation of nutrients.



The utilization of point-of-care clinical tools is recommended to educate, evaluate, treat, and monitor your patients' nutrient intakes.

The Basics of Nutrition for Optimal Function

Macronutrients

- Quality protein
- Balanced fats
- Fiber
- Complex carbohydrates

- Metabolic energy
- Amino acids for various physiological pathways
- Fatty acids for cellular health
- Fiber for motility and microbiome

Micronutrients

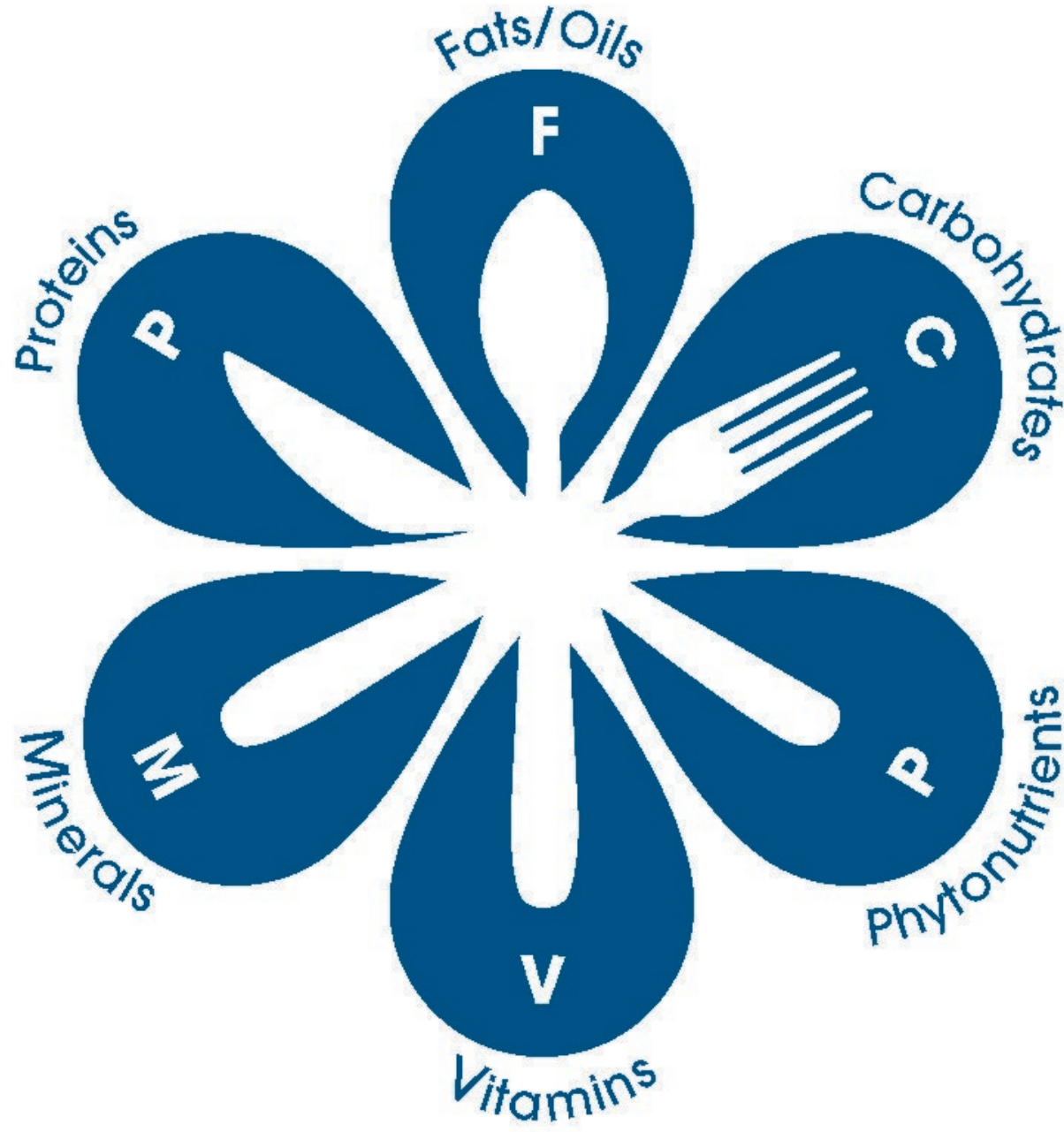
- Vitamins
- Minerals

- Cofactors for enzymes
- Antioxidant support

Phytonutrients

- Colorful pigments with functional properties

- Antioxidant support
- Anti-inflammatory
- Microbiome fortification



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Macronutrients and Micronutrients

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE